

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

**Pearson Level 3 Alternative Academic Qualification BTEC National
(Certificate/Extended Certificate)**

Sample assessment material

Time 2 hours

Paper reference **XXXXXX/XX**

Information Technology
UNIT 1: Information Technology Systems

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION A

Read the following scenario carefully. The questions in this section relate to this scenario.

Answer ALL questions. Write your answers in the spaces provided.

- 1 A mobile phone shop has shops where customers can buy phones and also have a website where customers can order phones online.

(a) The mobile phone shop collects customer data for online orders.

Give **two** external threats to unsecured data.

(2)

1

2

(b) Customers need to sign into their account using a password.

Describe the process used by a system to authenticate passwords.

(4)

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- Explain **two** validation methods the mobile phone shop could use in their data entry form to ensure data is accurate.

(4)

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- Explain **one** way a VPN ensures any data accessed remotely is secure during transmission.

(2)

- (e) The staff use emails to communicate with colleagues and customers.

Simple Mail Transfer Protocol (SMTP) and Internet Message Access Protocol (IMAP) are two protocols used to control the data transmission of emails.

Explain **one** feature of SMTP and **one** feature of IMAP in the transmission of emails.

(4)

SMTP

IMAP

(f) The mobile phone shop has decided to move to online-only sales.

Discuss the implications of the move to online-only sales for the mobile phone shop.

(6)

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(Total for Question 1 = 22 marks)

2 A company provides services to clients that include businesses and individuals.

The company has decided to move to bigger premises.

The company is concerned that during the move equipment is damaged and data is lost.

(a) State **two** impacts to the company if data is lost.

(2)

1

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(b) The company should ensure they design the computer workspace to minimise health risks.

Explain **two** factors the company should consider when they design the computer workspace in the new premises to minimise health risks.

(4)

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- (c) The company currently use a proprietary operating system and are considering replacing it with an open-source operating system.

Explain **two** benefits to the company of moving to an open-source operating system.

(4)

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- (d) A network operating system (NOS) is installed on the file server. Two functions of the network operating system are to manage the network and security.

Explain **one other** function of a NOS.

(3)

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(Total for Question 2 = 22 marks)

3 An online retail company sells clothing.

The retail company offers a 'click-and-collect' service.

Customers using the 'click-and-collect' service nominate a store to collect their purchases.

When customers place their online order, the stock availability in the warehouse is checked. Unavailable items cannot be ordered and the customer can select an alternative item.

Available items are despatched to the customer's nominated store.

Items not collected by the customer within 14 days are returned to the warehouse.

Emails are sent to the customer when:

- The item is despatched to the nominated store
- The item is ready for collection in-store
- The item has not been collected and has been at the store for 7 days.

(a) Draw a flowchart to show the process for the 'click-and-collect' service.

(6)

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- (b) A customer visits the website and sees a message stating there is a problem with the website's security certificate.

Explain **one** reason why websites use digital certificates.

(2)

- (c) The retail company uses a local area network (LAN) to store all its software and customer data.

The network topology installed is a 'bus'.

State **one other** network topology the company could have installed.

(1)

(d) Anti-virus software is installed on the retail company's IT system.

A virus alert is received on one of the computers stating a possible threat is detected and instructs the user to click a button to continue.

Describe how anti-virus software will protect the IT system against threats.

(4)

(e) The retail company stores all data on a server-based local area network (LAN).

The retail company is considering using cloud computing for all its IT systems. They believe this will be cost-effective.

Evaluate the cost implications for the retail company of a move to cloud computing.

You should consider:

- Benefits
- Drawbacks.

(9)

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(Total for Question 3 = 22 marks)

4 A manufacturing company produce a range of gaming accessories. The manufacturing company has an office and a factory on separate sites.

(a) The manufacturing company produce wireless and wired controllers.

Give **one** method to connect a controller to a console.

(1)

(b) The factory has both wired and wireless connections for its network.

Explain **one** benefit of using a wired connection.

(2)

(c) The manufacturing company has an acceptable use policy.

Evaluate the role of an acceptable use policy in protecting the company and their data.

(9)

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Handwriting practice area with 24 horizontal dotted lines.

- (d) The manufacturing company is considering introducing an emerging technology to improve services.

Evaluate whether the company should be more concerned with either the impact on:

- the performance of their IT systems
- the use of IT systems by employees.

(12)

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(Total for Question 4 = 24 marks)

TOTAL FOR SECTION A = 90 MARKS
TOTAL FOR PAPER = 90 MARKS

General marking guidance

- All students must receive the same treatment. Examiners must mark the first student in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Students must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark scheme, not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks if the student's response is not rewardable according to the mark scheme.
- Where judgement is required, a mark scheme will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark scheme to a student's response, a senior examiner should be consulted.
- Crossed out work should be marked unless the student has replaced it with an alternative response.
- Accept incorrect/phonetic spelling (as long as the term is recognisable) unless instructed otherwise.

Levels-Based Mark Scheme Guidance

Levels-based mark schemes (LBMS) have been designed to assess students' work holistically. They consist of two parts:

- **Indicative content:** Indicative content reflects content-related points that a student might make but is not an exhaustive list. Nor is it a model answer. Students may make some or none of the points included in the indicative content as its purpose is as a guide for the relevance and expectation of the responses. Students must be credited for any appropriate response.
- **Levels-based descriptors:** Each level is made up of a number of traits which when combined together articulate the quality of response that a student needs to demonstrate. The traits progress across the levels to demonstrate the different expectations of each level. When using a levels-based mark scheme, the 'best fit' approach should be used.

Applying the levels-based descriptors

- Examiners should take a 'best fit' approach to determining the mark.
- Examiners should first make a holistic judgement on which level most closely matches the student's response. Students will be placed in the level that best describes their answer. Answers can display characteristics from more than one level, and where this happens markers must use any additional guidance (e.g. weighting of traits) and their professional judgement to decide which level is most appropriate.
- The mark awarded within the level will be decided based on the quality of the answer and will be modified according to how securely all traits are displayed at that level.
- Marks will be awarded at the top of that level if the student has evidenced each of the descriptor traits securely.
- Where the response does not securely meet all traits, the marks should be awarded based on how closely the descriptor has been met.

Question Number	Answer	Mark
1a	Award one mark for each of the following, up to a maximum of two marks: • Viruses and other malware (1) • Unauthorised access / hackers (1) • Social engineering (1) Accept any other appropriate/alternative response	2

Question Number	Answer	Mark
1b	Award one mark for each appropriate point in a linked description up to a maximum of four marks. • database of usernames and passwords are created (1) • on entry password checked against the username (1) • access is allowed if the username and password matches (1) • re-enter password if there is no match / access is denied (1) • limited number of attempts are allowed (1) • if unsuccessful attempts exceed number allowed, account is locked (1) Accept any other appropriate/alternative response	4

Question Number	Answer	Mark
1c	Award one mark for identification and one additional mark for appropriate expansion, up to a maximum of two marks. Presence check (1) checks essential data fields have not been left blank (1) Format check (1) checks data entered (dob / post code / telephone numbers / price) use the correct format (date of birth / postcode / telephone number / price) using an input mask (1) Lookup table (1) looks up acceptable values in a table (product details against product number) (1) Length check (1) checks data entered is of the required length (for product number 8 numbers) (1) Multiple choice / Drop down list (1) using a list box to select from options for 'Customer title' (1) Type check (1) checks data entered is acceptable, e.g., text or number (1) Accept any other appropriate/alternative response	4

Question Number	Answer	Mark
1d	Award one mark for identification and one mark for an appropriate expansion point, up to a maximum of two marks. • A VPN ensures message integrity (1) by detecting any instances of interfering with transmitted messages (1) • authentication methods (1) by only allowing access from authorised devices (1) • VPN uses encryption (1), so data is unreadable if intercepted (1) Accept any other appropriate/alternative response	2

Question Number	Answer	Mark
1e	Award one mark for an identification, and one mark for an appropriate linked justification/expansion, up to a maximum of four marks. SMTP Manages the sending of email from server to server (1) to ensure it reaches the recipient (1) IMAP Stores emails on an email server (1) to allow access from multiple devices (1) Accept any other appropriate/alternative response	4

Question Number	Answer	Mark
1f	Answers will be credited according to the student's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Students discuss the implications of the move to online-only sales for the mobile phone shop. Mobile phone shop • cost implications ○ saving on individual showrooms (buildings / overheads / staff) ○ loss of customers / revenue from customers who prefer to visit the showroom ○ increased costs of additional staff to manage online sales and customer support services • potential for increased online sales • may need to set up an additional customer service facility to deal with increased queries • some data may be stored on computer systems in each of the showrooms ○ would need to ensure the data is backed up and transferred to central storage ○ removed / deleted from the showrooms systems • centralisation of all data ○ more manageable ○ fewer security issues ○ may lead to fewer technical staff	6

Mark scheme (award up to 6 marks) refer to the Levels-Based Mark Scheme Guidance for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates basic application of knowledge and understanding that is partially relevant to the context of the question and may consider only one side of the context. (AO2) • Demonstrates a basic analysis of the situation by superficially breaking down the different aspects into their component parts. (AO3a)
Level 2	3–4	• Demonstrates good application of knowledge and understanding that is relevant to the context of the question and considers both sides of the context. (AO2) • Demonstrates a good analysis of the situation by breaking down the different aspects into their component parts. (AO3a)
Level 3	5–6	• Demonstrates comprehensive application of knowledge and understanding that is consistently relevant to the context of the question and considers both sides of the context in a balanced way. (AO2) • Demonstrates a thorough analysis of the situation by comprehensively breaking down the different aspects into their component parts. (AO3a)

Question Number	Answer	Mark
2a	Award one mark for any of the following, up to a maximum of two marks: • Lost time in retrieving the data (1) • financial loss due to legal action (1) • loss of customers. (1) Accept any other appropriate/alternative response	2

Question Number	Answer	Mark
2b	Award one mark for an identification, and one mark for an appropriate linked justification/expansion, up to a maximum of four marks. • provision of ergonomic furniture/computer equipment / adjustable chairs designed to support users (1) to minimise back/posture problems (1) • ensure minimal light/glare is reflected on screens (1) by positioning screens appropriately / providing window blinds (1) • cables/wires should be securely tied up (1) to avoid risk of injury by tripping (1) • adjustable height/tilting of monitors (to eye-level) (1) so that users can minimise neck problems (1) • provision of ergonomic mice/keyboards (1) to reduce repetitive strain injury (RSI) (1) Accept any other appropriate/alternative response	4

Question Number	Answer	Mark
2c	Award one mark for an identification, and one mark for an appropriate linked justification/expansion, up to a maximum of four marks. • Access to source code (1) allows the company to meet specific needs (1) • The company will be able to work with other users to 'pinpoint' customisation (1) which would not be available with proprietary operating system manufacturers (1) • Bugs in the code are often noticed and corrected more quickly (1) because the users are not restricted by the manufacturer's procedures (1) • Can be more secure (1) because hackers tend to target proprietary software (1) • Ongoing support will be free of charge (1) because it is provided by other users/community rather than the manufacturer (1) Accept any other appropriate/alternative response	4

Question Number	Answer	Mark
2d	Award one mark for an identification, one mark for an appropriate linked justification, and one mark for a further expansion, up to a maximum of three marks. • Memory management (1) ○ allocates space/memory to files/programs/clipboard items that are running/open (1) and decides how much memory each process will get / it monitors each memory location's status to see if it is free/allocated (1) • Multi-tasking (1) ○ allocates each task a slice of processor time (1) and keeps track of where the user is in each task /without losing information (1) • Device drivers (1) ○ driver receives and translates signals so both operating system and applications software can understand (1) when a signal is received from the hardware OS it is held in a buffer / instruction put in a queue until resources are available (1) Accept any other appropriate/alternative response	3

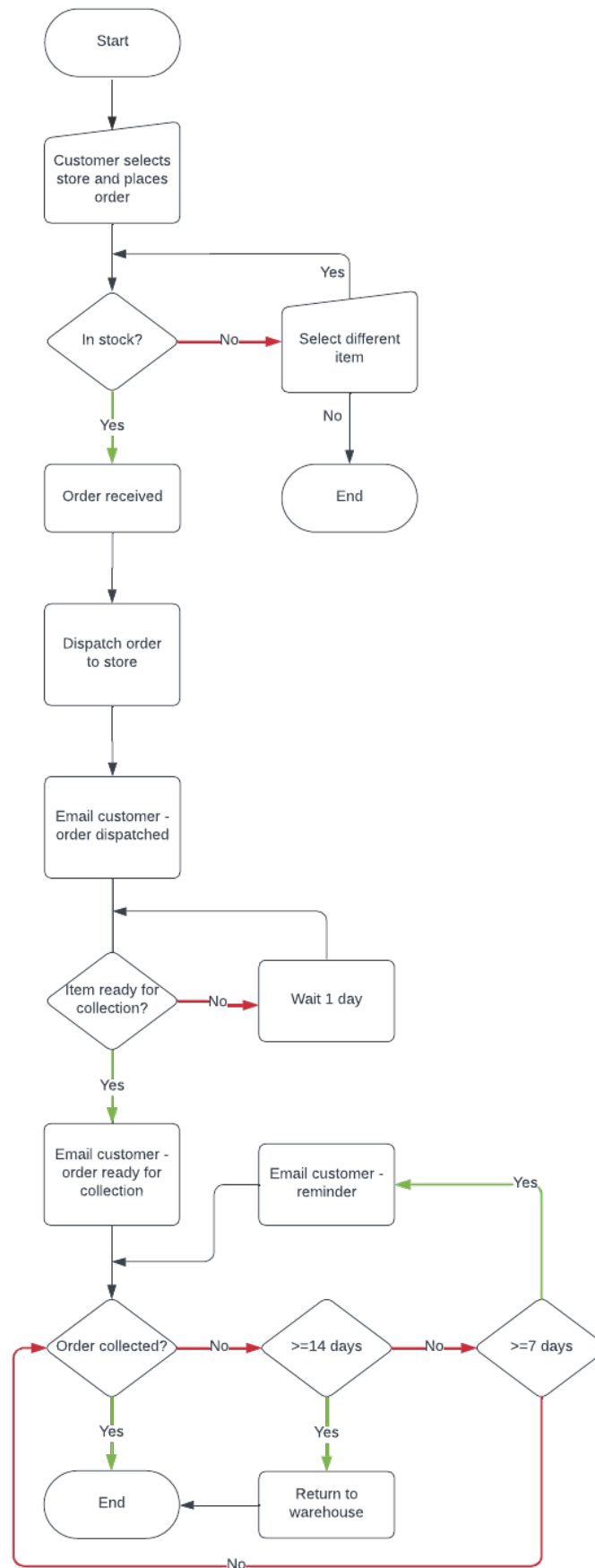
Question Number	Answer	Mark
2e	Answers will be credited according to the student's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Students evaluate the impact of updating all IT systems. They should consider the impact on the company and its employees. • Cost ○ will incur large initial costs for IT systems ○ may have to employ extra IT staff, initially and/or long term ○ staff training ○ may be able to re-coup some costs from selling used equipment • Disposal of old equipment ○ may need to bring in experts to ensure all data is removed and/or storage media destroyed ○ time and cost to dispose of hardware in an environmentally-friendly way • Implementation ○ direct changeover could impact the operations of the company if the new system does not work and the current system is obsolete ○ phased changeover would be more beneficial to allow the current staff to get used to the new system in phases ○ parallel changeover would be more beneficial because downtime would be kept to a minimum because the current system would still be running while the new one is being implemented. • Integration ○ other systems in use will need to be checked to ensure compatibility with the new system, e.g., any portable devices ○ files could be lost/corrupted during transfer to the new system • Productivity ○ loss of productivity during implementation ○ improve productivity / efficiency with the updated systems	9

	• Working practices ○ Changes to IT systems might have an impact on employees working from home, e.g., updating mobile technology, VPN costs • Staff training needs ○ cost ○ timing ○ in house / outsourced • Testing ○ must ensure that new system will work on all platforms ○ must ensure the new system will work with/ be suitable for current IT needs • Security ○ Of data during changeover ○ Setting up new systems ○ Up-to-date security software	
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Mark scheme (award up to 9 marks) refer to the Levels-Based Mark Scheme Guidance for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates basic application of knowledge and understanding that is partially relevant to the context of the question and may consider only one side of the context (AO2) • Demonstrates a basic analysis of the situation by superficially breaking down the different aspects into their component parts. (AO3a) • Demonstrates a basic evaluation which partially considers different factors/events and competing points, leading to a conclusion which is superficial or unsupported. (AO3b)
Level 2	4–6	• Demonstrates good application of knowledge and understanding that is relevant to the context of the question and considers both sides of the context. (AO2) • Demonstrates a good analysis of the situation by breaking down the different aspects into their component parts. (AO3a) • Demonstrates a good evaluation which considers different factors/events and competing points, leading to a conclusion which is partially supported. (AO3b)
Level 3	7–9	• Demonstrates comprehensive application of knowledge and understanding that is consistently relevant to the context of the question and considers both sides of the context in a balanced way. (AO2) • Demonstrates a thorough analysis of the situation by comprehensively breaking down the different aspects into their component parts. (AO3a) • Demonstrates a thorough evaluation which comprehensively considers different factors/events and competing points, leading to a conclusion which is well supported. (AO3b)

Question Number	Answer	Mark
3a	Logic of the problem (up to 4 marks only). • Order placed, stock availability in the warehouse is checked (1) • Unavailable items, process ends (1) • Available items are despatched to nominated store and email sent to customer (1) • Item is ready for collection in-store and email sent to customer (1) • Email to customer if not collected within 7 days (1) • Items not collected within 14 days are returned to the warehouse (1) Use of annotation regardless of logic (2 marks only) • Has start/begin AND stop/end symbols (1) • At least two decisions (regardless of shape) with exactly one input and exactly two outputs with 'Yes' and 'No' labels (1) • Accurate and correct symbols used throughout (1)	6



Question Number	Answer	Mark
3b	Award one mark for identification and one mark for an appropriate expansion point, up to a maximum of two marks. To verify identity of website/verify that a company owns a website (1) to ensure that the information is genuine/valid (1) User's browser software will allow access to sites with certificates (1) because the certificate verifies that website is authentic/legitimate (1) Provides encryption for communication (using encrypted attachment) (1) ensuring that (initial) data transmitted is protected (1) Do Not Accept – answers relating to showing website is secure e.g. Padlocks or https Accept any other appropriate/alternative response	2

Question Number	Answer	Mark
3c	Award one mark for any one of: • Ring (1) • Star (1)	1

Question Number	Answer	Mark
3d	Award one mark for each appropriate point in a linked description up to a maximum of four marks. • (Anti-virus software) monitors activity/scans computer/files (1) • compares (activity) against a database of definitions (1) • analyses the code/behaviour of the suspicious program (1) • Determines the nature of the threat (1) • Request actions from the user (1) • Give option to remove/quarantine the threat (1) • Log user actions to deal with similar future threats (1) Additional Guidance Do not accept 'removes the threat/virus' on its own Accept any other appropriate/alternative response	4

Question Number	Answer	Mark
3e	Answers will be credited according to the student's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Students evaluate the cost implications of a move to cloud computing. Students provide an evaluation of the decision, stating if they think the decision is appropriate or not. Their evaluation should be supported by relevant points which may include: Using a cloud computing system as an alternative to the current system will bring a range of cost advantages to the agency. In the traditional current system, they had to pay for the server and its installation, pay for and install applications and configure the system. There was also a range of ongoing costs relating to maintenance/updating etc. Using a cloud-based system the agency don't physically buy a server but access a part of a much larger server which is off-site away from the business. This is likely to be much cheaper in the end, although there will be costs incurred with running the system. <u>Costs will be saved</u> as there is no need to pay for: • expensive hardware on an ongoing basis – either to update or to add capacity • updating software • ensuring up-to-date security • technical staff to maintain the system / deal with user issues etc • software, including OS and applications. • unused server capacity, which would be fixed under current system, but now can be scaled up and down to fit the needs of the business • access to up-to-date technology, without having to make huge investments • robust disaster recovery, which could be very costly in terms of expertise. • bespoke application packages or other commercially available software, cloud business applications are offered by suppliers and are much cheaper • VPN for home-based workers to access server <u>Costs will be incurred for:</u>	9

	• access to the new system. May be subscription based or pay as you go model, which will give them authorisation to use the software for a period of time and pay only for the software that they are using • Included in the cost will be a payment for support/staff etc. • server capacity, which will be dependent on what they are using mobile apps for staff, as required, allowing use on a variety of devices to access software & data	
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Mark scheme (award up to 9 marks) refer to the Levels-Based Mark Scheme Guidance for how to apply levels-based mark schemes*.

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Level 3	7-9	• Demonstrates comprehensive application of knowledge and understanding that is consistently relevant to the context of the question and considers both sides of the context in a balanced way. (AO2) • Demonstrates a thorough analysis of the situation by comprehensively breaking down the different aspects into their component parts. (AO3a) • Demonstrates a thorough evaluation which comprehensively considers different factors/events and competing points, leading to a conclusion which is well supported. (AO3b)

Question Number	Answer	Mark
4a	Award one mark for any of the following: • Bluetooth (1) • USB (1) Accept any other appropriate/alternative response	1

Question Number	Answer	Mark
4b	Award one mark for identification and one mark for an appropriate expansion point up to a maximum of two marks. • Signal is reliable/stable (1) because it is not affected by interference/walls/other devices (1) • Signal does not deteriorate over distance (1) which would reduce the need for boosters (1) • More secure (connection) (1) because it is not as easily intercepted (1) • Faster transmission speeds (1) that allows large files to download faster (1) Accept any other appropriate/alternative response	2

Question Number	Answer	Mark
4c	Answers will be credited according to the student's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Students evaluate the role of an acceptable use policy to protect the company and their data. • Protects the manufacturing company's users and data • Minimises the risk by enforcing the expected behaviours of all staff when using IT systems and resources • Ensures all staff understand their responsibilities for the appropriate use of IT systems and resources ◦ IT equipment includes all hardware, software, social media, internal and external emails, internet and intranet Employees must: • Agree to comply with the Acceptable Use Policy (AUP) • Any breaches may result in disciplinary action • Act responsibly and professionally and be responsible for their own actions • Ensure they use the IT equipment in line with the manufacturing company's security policy • Report any breach of the AUP to the relevant team • Ensure usernames and passwords are used appropriately and securely ◦ Not written down or stored in shared folders ◦ Not allowing employees to log on to the systems using a colleague's username and password ◦ Log out of IT systems when not in use Company must: • Monitor the business and personal use of the IT systems • Ensure staff have the necessary training to ensure they are aware of security procedures Protecting data: • The manufacturing company and its employees have legal responsibility to protect personal/sensitive data • Not to access personal data unless it is appropriate to do so for the job role Using IT systems: • Employees are accountable when using online systems and should not damage the reputation of the company ◦ Not downloading illegal music/video/software ◦ Not bullying or harassing ◦ Not posting offensive material that could embarrass or compromise the company ◦ Not breaking company confidentiality online	9

Mark scheme (award up to 9 marks) refer to the Levels-Based Mark Scheme Guidance for how to apply levels-based mark schemes*.

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Question Number	Answer	Mark
4d	Answers will be credited according to the student's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all the indicative content but should be rewarded for other relevant answers. Students evaluate the implications for the performance of IT systems and the use of IT systems by employees. Emerging technologies can be hardware, software based or a combination. The focus of responses may relate to any/all of these. Impact on employees: The company should be concerned with the overall performance of employees when using the emerging technology in the short and long term based on: • ease of use: ○ whether initial/ongoing training on the emerging technology would be required and how long this would take/how expensive it would be ○ whether investment in time/money for training would be offset in the longer term ○ whether the emerging technology would speed up/slow down use of IT by employees. • performance: ○ whether the emerging technology would be faster than the previous systems ○ whether employees can be more productive ○ whether introducing the emerging technology may replace some of the processes undertaken by employees. • availability: ○ whether the emerging technology would be available to all employees ○ whether certain departments/processes/employees would benefit more from the emerging technology than others. • accessibility: ○ whether the emerging technology places any barriers on employees engaging with it/performing their duties/responsibilities. Impact on performance: need to ensure that the emerging technology is compatible across different platforms. This includes hardware, operating systems and other application software, communication software, security software including:	12

	• current hardware devices. These may vary, could be laptops, tablets, gaming devices. • peripherals may not work with new/existing software. • operating systems. If not compatible, then the manufacturing company will need to consider the implications of replacing / upgrading the device. • application software. Depending on type of emerging technology may need to work alongside other applications and compatibility is essential. • security software. Must ensure that the emerging technology does not cause security issues. • who is going to install/maintain the technology – in-house / external agency and how will this impact performance? • must ensure that the emerging does not impact on the security of current systems that impact on performance. • does the technology need all/some system(s) to be closed when installing, if so will need to carefully consider when the installation is going to take place? • has the technology been fully tested or are issues likely to arise which will impact on other aspects of the system(s).	
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Mark scheme (award up to 12 marks) refer to the Levels-Based Mark Scheme Guidance for how to apply levels-based mark schemes*. AO2 descriptors carry double weighting

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates basic application of knowledge and understanding that is partially relevant to the context of the question and may consider only one side of the context (AO2) • Demonstrates a basic analysis of the situation by superficially breaking down the different aspects into their component parts. (AO3a) • Demonstrates a basic evaluation which partially considers different factors/events and competing points, leading to a conclusion which is superficial or unsupported. (AO3b)
Level 2	5–8	• Demonstrates good application of knowledge and understanding that is relevant to the context of the question and considers both sides of the context. (AO2) • Demonstrates a good analysis of the situation by breaking down the different aspects into their component parts. (AO3a) • Demonstrates a good evaluation which considers different factors/events and competing points, leading to a conclusion which is partially supported. (AO3b)
Level 3	9–12	• Demonstrates comprehensive application of knowledge and understanding that is consistently relevant to the context of the question and considers both sides of the context in a balanced way. (AO2) • Demonstrates a thorough analysis of the situation by comprehensively breaking down the different aspects into their component parts. (AO3a) • Demonstrates a thorough evaluation which comprehensively considers different factors/events and competing points, leading to a conclusion which is well supported. (AO3b)

July 2024

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Pearson BTEC Level 3 Nationals Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Friday 10 January 2025

Morning (Time: 2 hours)

Paper reference	31760H
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Information Technology
UNIT 1: Information Technology Systems

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

Answer ALL questions. Write your answers in the spaces provided.

1 Designs for Us is a company that creates logos and advertising materials.

(a) State **two** input devices the company's designers could use to create logos.

(2)

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(b) (i) The designers store files on a central server.

Explain **two** benefits to the designers of storing files on the central server.

(4)

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(ii) Explain **two** benefits to the company of storing files on the central server.

(4)

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P 8 1 1 9 7 A 0 3 2 4

(c) The designers mainly work remotely. They use a virtual private network (VPN) to connect to the company network.

Explain **two** benefits to the designers of using a VPN.

(4)

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(d) The designers save their images using different file formats.

Discuss the features of **two** file formats the designers could use to save their image files.

(8)

(Total for Question 1 = 22 marks)



P 8 1 1 9 7 A 0 5 2 4

- 2 A sixth form college has a local area network (LAN) that students use to access resources and learning materials.

The students save their work to the network when in college or at home.

- (a) Utility software is installed on the network.
(i) Anti-virus software is one type of utility software.

Explain the purpose of **two other** types of utility software.

(4)

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(ii) Describe how the anti-virus software works to reduce the risk of viruses infecting the college network.

(4)

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- (b) Staff enter the server room using a keypad entry system. The four-digit code is changed each week.

The college is updating to an ID card entry system.

Explain **two** ways an ID card entry system makes access to the server room more secure than a keypad entry system.

(4)

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(c) Students are required to log into the college network.

Describe the login authentication process.

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(d) Discuss the moral and ethical factors for staff and students when using IT in the classroom.

You need to consider:

- privacy
- online behaviour and netiquette
- acceptable use.

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(Total for Question 2 = 24 marks)



P 8 1 1 9 7 A 0 1 1 2 4

- 3 CG is a company that sells electrical products. It has a chain of stores and a warehouse.

Each store has a stockroom with a desktop PC and printer.

Electronic Point of Sale (EPOS) terminals are used by sales staff to complete sales and process customer payments.

The EPOS system is connected to the desktop PC in the stockroom. When a sale is made stock levels are updated automatically on the desktop PC.

At the end of each day, a stock request is sent to the database server in the warehouse from the desktop PC in the stockroom. Stock requests are printed from a desktop PC in the company warehouse.

- (a) Draw a diagram to show how a store and the warehouse interact.

The diagram should include devices and systems that are used, including:

- desktop PCs
- EPOS terminals
- payment card readers
- printers
- database server
- devices to be connected and the connection type(s) used
- annotations.

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Use this space to draw your diagram for Q3a.

(6)



(b) CG uses digital certificates and protocols to protect its data and systems.

Explain **two other** ways CG can protect its data and systems.

(4)

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- (c) CG has a secure website that uses a digital certificate where customers can place online orders.

Explain **two** reasons why websites use digital certificates.

(4)

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(d) Discuss the security protocols websites use for data transmission.

(8)

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(Total for Question 3 = 22 marks)



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4 Clothes for Us is a company that designs clothes for high-street shops.

The company is considering moving its data to a cloud-based storage system.

- (a) Discuss the positive and negative impacts to the company and staff of moving to cloud-based storage.

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(b) Clothes for Us has a company network. The company’s designers use the network to work collaboratively.

Evaluate the features of the company network that will allow the designers to work collaboratively.

(12)

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(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 90 MARKS





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MS - An exam paper

Information Technology in a Global Society SL (Leyton Sixth Form College)



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Mark Scheme (Results)

January 2025

Pearson BTEC Nationals
In IT (31760H)
Unit 1: Information Technology Systems

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January 2025

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Unit 1: Information Technology Systems

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if a candidate's response is not worthy of credit according to the mark scheme.
- Where some judgment is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt about applying the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Phonetic spelling should be accepted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

BTEC Next Generation Mark Scheme Template

IT Unit 1 January 2025

Question Number	Answer	Mark
1a	Any two from: Mouse / trackerball / joystick (1) Stylus / pen (1) Digitiser (1) Scanner (1) Keyboard (1) Drawing/graphics tablet (1) Touch screen (1) Camera (1) Accept any other relevant input device.	2

Question Number	Answer	Mark
1bi	Award up to two marks for each of two linked explanations, from: • enables the designers to have a more flexible working pattern (1) because designers can access their files from home / not restricted to one workstation (1) • information is easily accessible (1) because all files are stored in one place (1) • designers can collaborate on a project (1) because they can share/access the same files (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
1bii	Award up to two marks for each of two linked explanations, from: • improved data security (1) because it makes it easier to apply security measures / easier to monitor user activity (1) • easier to backup data (1) because the company can schedule automatic backups / because all files are stored in one place (1) • cost effective (1) because the company does not have to purchase/maintain additional equipment/storage (1) • scalability (1) because the company can update/expand the server when required (1) Accept any other relevant phrasing/wording Allow: examples of security measures	4

Question Number	Answer	Mark
1c	Award up to two marks for each of two linked explanations, from: • the connection will be secure/encrypted (1) prevents unauthorised access to the device (1) • improved privacy when using shared computers (1) as all traffic is kept inside the secure VPN 'tunnel' (1) • avoids bandwidth throttling (1) because it hides the activity from their (Internet Service Provider) ISP (1)	4

Question Number	Indicative content
1d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the features of suitable image file formats. <u>JPEG</u> • lossy compression • suitable for sharing images • universal browser and OS support • fairly low file size • lossy image compression might lead to poor text readability • supports 24-bit colour <u>GIF</u> • lossless compression • limited palette of colours (limits to 8 bits per pixel) • animation support • low file size • 8-bit limitation leads to limited image quality • supported by all major browsers and OS and image editors <u>PNG</u> • lossless compression • maintains detail and contrast between colours • high quality • larger files can slow down the website if overused (especially high-resolution images) • supported by all major browsers and OS • transports transparency <u>TIFF</u> • supported by most digital platforms and devices • commonly used for storing and editing images • typically used as a lossless image format • high quality files perfect for storing or printing publishing • large file size due to no compression • limited browser support without add-ons or file extensions • mainly used as an export format for professional image editing and publishing tools <u>BMP</u> • lossless • maps individual pixels with no compression • can become extremely large file sizes but retains high quality and remain the same when you upload them • supported by all major browsers • most default OS image editor

	<u>SVG</u> • 2-dimensional vector images • web-friendly format, can be read by search engines using keywords • can be resized without losing quality • small file sizes (XML) • can be edited with text editors/drawing software • supports animation and interactivity
--	---

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer	Mark
2ai	Award one mark for identification and two additional marks for appropriate expansions. • File management software (1) is used to help browse, search, and organise files on the system quickly (1) • Disk cleanup tool/uninstaller (1) is used to free up disk space by removing unnecessary/temporary files (1) • Backup utility (1) is used to create a copy of stored data / can automatically back up your data / to ensure lost data/files can be retrieved (1) • File compression tools (1) are used to reduce the size of files for storage / files that are sent as email attachments (1) • Disk defragmenter (1) is used to rearrange fragmented data into contiguous blocks to reduce read/write times (1) • Firewall software (1) because it monitors incoming and outgoing traffic / blocks unauthorised access (1) • VPN software (1) because it encrypts the network connection / sends data packets through a secure tunnel (1) • Encryption software (1) so that data is unreadable / can only be understood by authorised users (1) Accept any other relevant phrasing/wording Do Not Accept: 'scrambled' for encryption in this instance.	4

Question Number	Answer	Mark
2aii	A description of how the college's antivirus software will protect the network. To include any four from: • (Anti-virus software) scans computer/files (1) • compares threats detected against a database of definitions (1) • analyses the code/behaviour of the suspicious program (1) • determines the nature of the threat (1) • alerts the user (1) • give option to / will remove/quarantine the threat (1) • log user actions to deal with similar future threats (1)	4

Question Number	Answer	Mark
2b	Award up to two marks for each of two linked explanations, from: • An ID card system does not require the staff to enter a code (1) therefore unauthorised users / shoulder surfers could not copy the number (1) • A four-digit code can easily be leaked to (numerous people) (1) whereas the ID card can only be passed to one person (at a time) (1) • the college can (quickly) deactivate the card (1) because the member of staff would report their card has been lost or stolen promptly (notice it has gone) (1) • ID cards cannot be copied (easily) (1) because they require specialist equipment / have in built 'technology' (1) • An ID card is unique to the card holder (1) to ensure that the user of the card is the person authorised (to access the server room) (1) Accept any other relevant phrasing/wording Accept mix and match if the response makes sense.	4

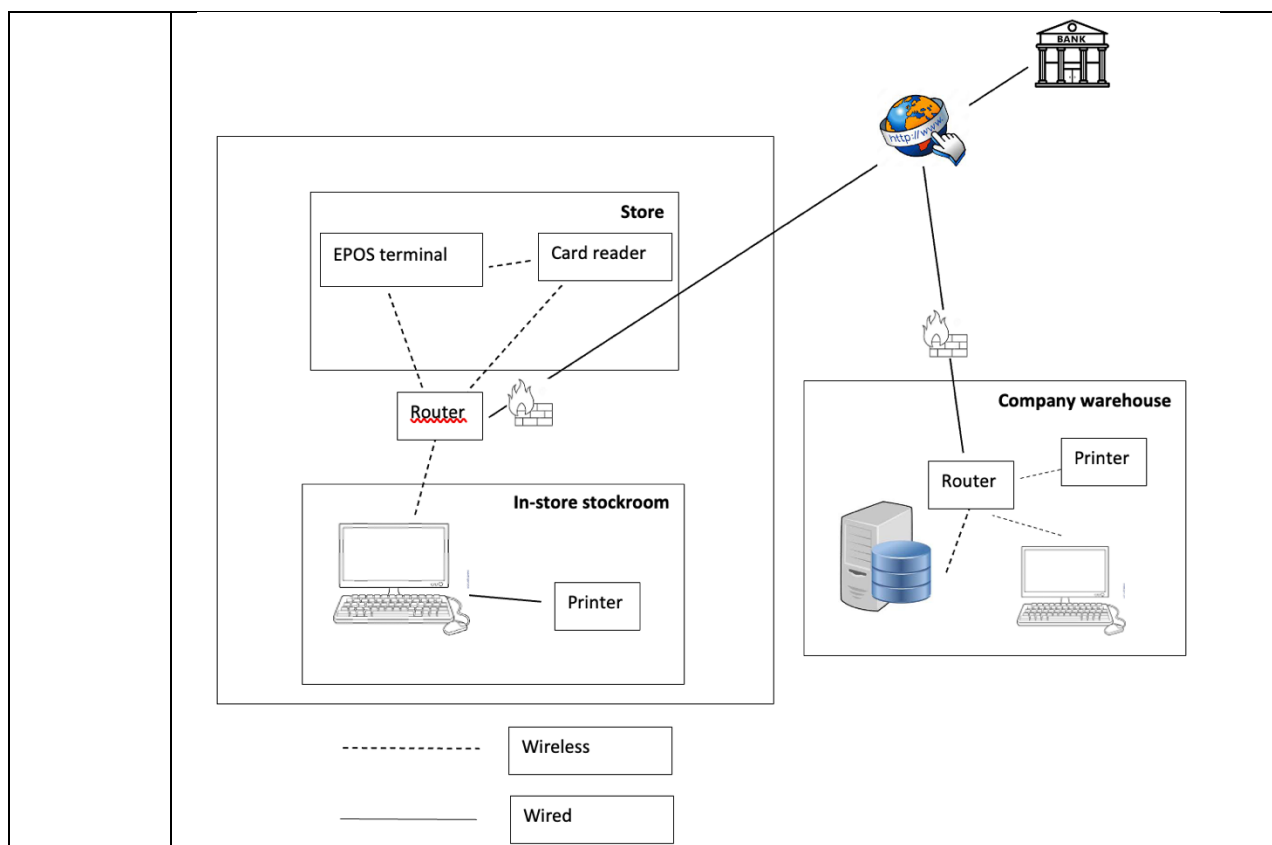
Question Number	Answer	Mark
2c	Award one mark for each point up to a maximum of four marks. • input student's username and password (1) • system checks username and password against a database of saved usernames/passwords (1) • grant access if a match (1) • display error message if no match / asked to re-enter details (1) • system allows a limited number of attempts (1) • account is locked when unsuccessful attempts exceed number of attempts (1) Accept any other relevant phrasing/wording Allow: 'email' for 'username'	4

Question Number	Indicative content
2d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the moral and ethical factors for staff and students when using IT in the classroom. <u>Privacy</u> • where portable devices such as phones are used and what is being said to avoid eavesdropping/passing on of private/sensitive data • who can see computer screen - switch off monitor/put machine into sleep mode when not using/ moved away from PC • who can see contents of documents/emails • make sure passwords conform to college standards / are not written down / not shared with other staff / students • privacy policies of websites. <u>Online behaviour and netiquette</u> • treating others with respect / use appropriate language / do not engage in cyber bullying • be professional when dealing with others • only visit trusted websites • be aware of malware and phishing – do not open links / attachments in emails from unknown senders • do not upload / post illegal / offensive materials that could cause offence to others • breach copyright laws when downloading materials • respect the privacy of others • consider tone when posting comments • do not shout using capitals <u>Acceptable Use</u> • internet use should be restricted to research for college purposes • file permissions should be enforced so staff and students do not access other people's folders without permission • IT systems (internet, email and mobile technologies) should only be used for college purposes • downloading / installing software onto college systems should be prohibited • take care using IT equipment at all times – do not pull any cables out / bypass web filters • do not access social media websites unless authorised • 'think before you print' – only print out what is essential, saving both paper and energy • switch devices off or into sleep mode when not in use • set power saving features • ensure that 'redundant' materials are correctly recycled/disposed of e.g. printer cartridges, can they be refilled

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
3a	A diagram of a potential system/integration of IT systems and connections used to meet the requirements of the scenario. Example response – note this is indicative only. Credit should be given for a diagram that meets the requirements of the given problem. Diagram may include: Devices: • Stock room – Desktop PC, and printer • Store – EPOS terminal, card reader • Warehouse – database server, desktop PC, printer, switch, and router Connection types and users: EPOS terminals • EPOS terminal to desktop PC – wireless / wired • EPOS terminal to internet – wired • Card reader to customer’s bank – wired Store stock room • Desktop PC to EPOS system – wireless / wired • Desktop PC to warehouse – wired Company warehouse • Internet to database server – wired • Internet to desktop PC – wired • Desktop PC to database server – wired/wireless • Desktop PC to printer – wired/wireless Annotations: The diagram should include annotations for devices and systems as well as the connection types used.



Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-2	- Diagram provides partial coverage of devices. - Limited connections are shown - Diagram includes some annotations of the connection types used.
Level 2	3-4	- Diagram provides coverage of mostly appropriate devices. - Connections are shown which meet most requirements of the scenario. - Diagram includes some annotations of connection types used to meet the majority of the requirements of the scenario, but these may not always be the most efficient.
Level 3	5-6	- Diagram provides thorough coverage of appropriate devices which fully meet the requirements of the scenario. - Connections are shown which fully meet the requirements of the scenario. - Diagram includes detailed and accurate annotations of devices and connection types used to meet the requirements of the scenario.

Question Number	Answer	Mark
3b	Award up to two marks for each of two linked explanations, from: • file permissions (1) because these control who can access specified files and what actions they are permitted to perform (1) • access levels (1) allows users to access specific data and systems (as defined in their job role) / designates the type of access (read only or read/write) • backup and recovery procedures (1) is the process of duplicating data and storing it in a secure place in case of loss or damage so it can be restored at a later date (1) • passwords (1) are used to allow authorised staff access to data and systems (1) • physical access control / biometrics (1) are used to determine which staff are authorised to enter areas of the warehouse (1) • anti-virus software (1) which scans for viruses (and alert the user to either remove or quarantine the threats) (1) • firewalls (1) which monitors incoming and outgoing traffic / blocks anything suspicious (1) • encryption (1) this protects data and systems by scrambling data so it is unreadable (1) • VPN (1) creates a secure tunnel that protects data when sending to external sources (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
3c	Award up to two marks for each of two linked explanations, from: • to verify identity of website / verify that a company owns a website (1) which proves that the information is genuine/valid/legitimate (1) • protect data integrity (1) to ensure that data has not been tampered with (1) • verifies that the website is authentic/legitimate/official (1) so that antivirus software will allow access to the site (1) • indicates the website can be trusted/reassures customers (1) allowing customers to know their personal/payment details are safe (1) • digital certificates encrypt the communication (between the web browser and website) (1) to protect customers' personal/payment details (1) Do Not Accept – answers relating to showing website is secure eg Padlocks or https Accept any other relevant phrasing/wording	4

Question Number	Indicative content
3d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners will discuss the security protocol websites use for data transmission. <u>HTTPs (Hypertext Transfer Protocol secure)</u> • is a secure version of HTTP • encrypt users HTTP requests and webpages to increase security of data transfer • protects against eavesdroppers/man in the middle attacks <u>SSL (Secure Socket Layer)</u> • Provides privacy and data integrity for internet communications • Encrypts the link between a website and browser / two servers • Ensures data transmitted between a client and server remains private • Prevents hackers from reading any data transmitted • The client is able to authenticate the identity of the server <u>TLS (Transport Layer Secure)</u> • Provides privacy and data integrity for internet communication • More secure version of SSL • Ensures the secure delivery of data over the internet / avoids eavesdropping • Can be used for other applications such as emails / file transfers / video conferencing / VOIP

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the impact on the company and its staff of moving to cloud storage. Positive impacts <u>Company</u> • Cost: no need to pay for technical staff, either in-house or on a consultancy basis, to provide a backup system this will be done automatically. • Storage capacity can be scaled up and down to fit the needs of the business. This has the added benefit of being better for the environment than running a server with excess capacity. • Will give Clothes for Us robust disaster recovery, which would otherwise be unlikely due to lack of finance and expertise. • Reduces the implications/security risks of lost or stolen laptops and other mobile devices. Data stored on the cloud is automatically backed up and therefore instantly retrievable. • Improved security arises from the reduction in the need to send files via external methods such as email. <u>Staff</u> • Because the system can be used anywhere with an internet connection, staff will be able to access the files/folders when they are out of the office or working remotely. • Many suppliers provide mobile apps that will allow staff to access data on a range of devices. • Some staff, e.g. office staff, may be able to take advantage of the system and work from home. • Because all files are stored centrally everyone sees the latest/same version leading to improved collaboration. Reducing the problems of conflicting file content, formats and titles. • Can have their own logins and access their files from their own personal storage area Negative impacts <u>Company</u> • Introducing a third party may have security implications for Clothes for Us if the 3rd party does not have robust security procedures in places for backup and recovery and data transfer • Training of staff would be time consuming and costly to the company and could result in downtime while training is taking place • Cost: ongoing subscription payments to the 3rd party provider, expensive if large amounts of storage is required for the designs • Downtime: designers could not complete designs in time, which could result in other companies having a competitive advantage

	<u>Staff</u> • Must have an internet connection, could be a disadvantage to staff out of the office. • Impact of downtime – staff will not be able to access their files if the cloud storage is down or no internet, which could lead to deadlines being missed and payment could be withheld • Staff might not want all files stored on a shared area because other people could copy their designs and use for their own purposes • Frustrating for staff to get used to new technology which can affect their work productivity
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Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5–7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8–10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners evaluate the features of the network that designers could use to work collaboratively. Shared area: • Can be set up to centrally store designs so designers can access the shared area remotely • Allows all designers to access/edit documents from a central storage point • Version control can be maintained to ensure designers are working on the most current version • Applications running on the network can enable designers to work in real time Management of groups: • Only authorised members of the organisation will have access to the share area • Designers will have access to the same software to create their designs • Ensure no compatibility issues with the designers using devices remotely

Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited evaluation which contains generic or descriptive points • Little or no conclusion
Level 2	5-8	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed evaluation which considers some different competing points • There may be a conclusion which is partially supported
Level 3	9-12	• Demonstrates mostly accurate and detailed knowledge and understanding • There may be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical evaluation which clearly considers different aspects and competing points in detail • There is a conclusion that is supported

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Welsh Assembly Government

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Candidate surname						Other names					
Centre Number						Learner Registration Number					
Pearson BTEC Level 3 Nationals Extended Certificate, Foundation Diploma, Diploma, Extended Diploma											
Friday 12 January 2024											
Morning (Time: 2 hours)						Paper reference		31760H			
Information Technology UNIT 1: Information Technology Systems											
You do not need any other materials.										Total Marks	

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1 JRC is a theatre company that uses IT systems.

(a) The Health and Safety (Display Screen Equipment) regulation is one piece of legislation used to protect users and data when using IT systems.

(i) Give **two other** pieces of legislation that JRC should consider when using IT systems.

(2)

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(ii) Explain **two** ways to reduce health and safety issues when using IT systems.

(4)

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When viewing stored card information the card number is partially replaced by asterisks.

Explain **one** reason why the card number is displayed as asterisks.



(d) JRC has decided to use e-ticketing rather than printing paper tickets.

Explain **two** benefits to JRC of using e-tickets.

(4)

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(e) Discuss the benefits and drawbacks to the customer of purchasing tickets online.

(8)

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(Total for Question 1 = 24 marks)



2 Poppy is a long-distance runner. She is preparing to compete in the world championships.

(a) Poppy uses a smartwatch to monitor her heart rate when running. She uses wireless connectivity to transfer the data from her heart monitor to her tablet computer.

(i) Explain **two** benefits to Poppy of using wireless connectivity to transfer data from her heart monitor to her tablet computer.

(4)

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(ii) Explain **two** threats to the data stored on Poppy's tablet computer.

(4)

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- (b) Poppy sends data from her tablet computer to her coaching team. The data is encrypted to ensure it is secure.

Describe how encryption works to keep the data secure during transmission.

(4)



(c) Poppy connects her tablet computer to her other devices using a personal area network (PAN).

Explain **two** reasons why Poppy would use a PAN to connect her devices.

(4)

1

2



(d) Poppy's coaching team needs to keep track of her location when she is running.

They do this using a global positioning system (GPS) tracking device.

Location is one example of information that can be calculated using GPS.

Discuss **other** examples of information that can be calculated using GPS.

(6)

(Total for Question 2 = 22 marks)



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QUESTION 3 BEGINS ON THE NEXT PAGE.



3 A motor vehicle garage provides services and MOTs.

The garage has a small reception area.

The reception area includes a:

- wireless laptop
- database server to store customer, service and MOT records
- card reader for customers to make secure payments
- printer for any paper invoices.

(a) Draw a diagram to show the integration of systems used in the garage.

The diagram should include devices and systems that are used, including:

- laptop
- database server
- card reader
- printer
- internet access
- connection type(s)
- annotations.



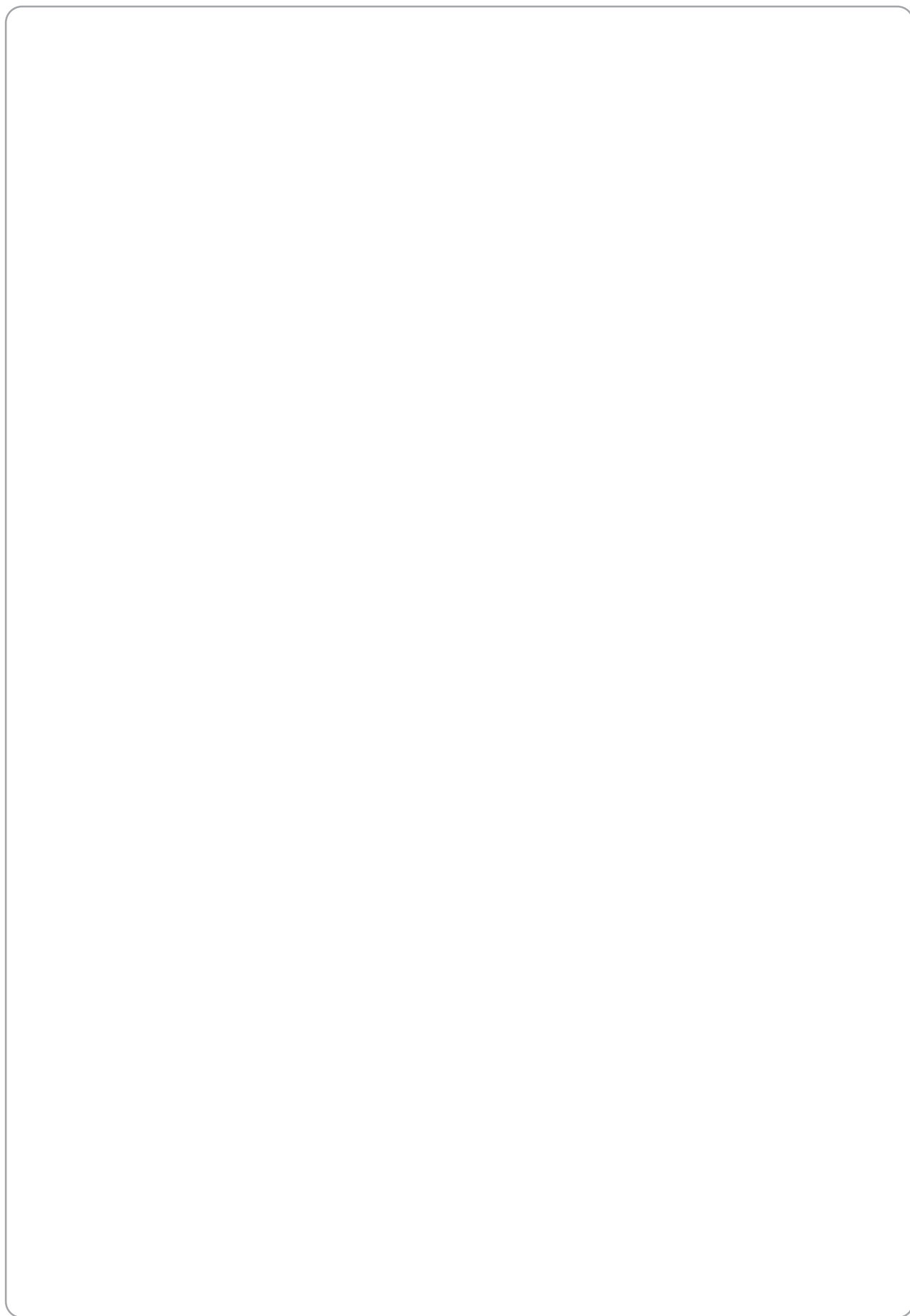
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Use this space to draw your diagram for Q3a.

(6)



(b) The card reader allows contactless payments so customers can pay by card or smartphone.

(i) Explain **two** benefits to the customer of making contactless payments.

(4)

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- (ii) The owner is considering introducing a self check-in system for customers to use when arriving at the garage.

Explain **two** characteristics that need to be considered when designing the user interface for the self check-in system.

(4)

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- (c) The garage stores all data on a database server. The owner is concerned there is no backup if the server fails.

The owner is considering moving all the data to an online storage system.

Discuss factors that affect the use and selection of online systems for storing data.

(8)



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(Total for Question 3 = 22 marks)



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4 A large advertising company is considering upgrading its network operating system.

The company will need to purchase new software licences for the upgrade.

(a) Discuss the factors that the company will need to consider before deciding to upgrade its network operating system.

You should consider:

- costs, other than licences
- implementation
- training
- compatibility
- support.

(10)



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P 7 5 7 9 8 A 0 1 9 2 4

- (b) The advertising company uses computers. Computers can have a significant impact on the environment.

Analyse ways the company and the staff can reduce the impact of company computers on the environment.

(12)



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TOTAL FOR PAPER = 90 MARKS



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Candidate surname						Other names					
Centre Number						Learner Registration Number					
Pearson BTEC Level 3 Nationals Extended Certificate, Foundation Diploma, Diploma, Extended Diploma											
Friday 9 June 2023											
Morning (Time: 2 hours)						Paper reference		31760H			
Information Technology UNIT 1: Information Technology Systems											
You do not need any other materials.										Total Marks	

Instructions

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- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1 Media students in a school create information and event materials.

(a) The media students use word processing software to create the materials.

(i) Identify **two other** types of software the media students could use to produce the materials.

(2)

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(ii) Explain **two** benefits of using word processing software to create the materials.

(4)

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- (b) The students at the school use the internet to research projects. The headteacher is concerned about malware.

One type of malware is a virus that is designed to spread from one computer to another by replicating itself.

Explain **two other** types of malware.

(4)

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- (c) The headteacher wants to use an online web form to gather data from students and their parents.

Explain **three** implications that should be considered when designing the web form.

(6)

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3



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This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice. There are no margins, text, or other markings on the page.



2 Luca runs a family business from home. His two children study at a college.

(a) Luca has purchased laptops for the family that do not have any software installed.

Explain the terms 'application software' and 'systems software'.

(4)

Application software

System software



(b) Luca is considering installing open-source system software.

- (i) Explain **two** advantages of installing open-source system software rather than proprietary system software.

(4)

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- (ii) Explain **one** disadvantage of installing open-source system software.

(2)

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(c) Luca wants to ensure the data on the laptops is protected and secure.

(i) One way Luca could protect the data is to use antivirus software.

Describe how the features of antivirus software will protect the data on the laptops.

(4)



- (ii) One threat to the security of data is phishing, which tricks victims into handing over sensitive information.

Discuss **other** threats to data.

(8)

(Total for Question 2 = 22 marks)



3 Craig is the manager of a leisure centre.

A new network is to be installed.

The leisure centre includes:

- a storeroom
- a reception area for the staff to take bookings
- an office where all administration work is undertaken
- a members' area with wireless laptops for use by members.

Craig and his staff have access to tablets that they use for personal training sessions and classes.

- (a) Draw a diagram to show the integration of systems and software used in this process.

The diagram should include devices and systems that are used, including:

- a server in the storeroom
- 2 desktop PCs in the reception area with wired connections
- 2 desktop PCs and a printer in the office with wired connections
- 3 laptops in the members' area with wireless connection to the internet
- 3 tablets with wireless connection
- connection type(s) used
- annotations.



Use this space to draw your diagram for Q3a.

(6)

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P 7 0 5 3 2 A 0 1 1 2 0

(b) A single-user, multitasking operating system is installed on the tablets.

Describe **three other** types of operating system.

(6)

- 1
- 2
- 3



(c) Members can pay for classes via the leisure centre’s website using their credit or debit card.

Describe how SSL/TLS protocols are used to secure this process.

(4)

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(d) Many factors influence the leisure centre's choice for the new network.

Purchase cost and **security** are two of these factors.

Discuss **other** factors that should be taken into consideration.

(8)



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(Total for Question 3 = 24 marks)



P 7 0 5 3 2 A 0 1 5 2 0

4 Seren Health Centre has a computerised patient details system.

The health centre uses processes to protect its system and patients' data.

One process it uses is username and password authentication.

(a) Discuss **other** processes the health centre could use to protect its system and patients' data.

You need to consider:

- file permissions
- access levels
- backup and recovery procedures
- physical access control.

(10)

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- (b) The health centre uses social media to keep patients up-to-date on important information about the health centre.

Evaluate the advantages and disadvantages of the use of social media in the healthcare sector.

(12)



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(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 90 MARKS



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Mark Scheme

May 2023

Pearson BTEC Level 3 - IT

Unit 1: Information Technology Systems
31760H

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Summer 2023

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

BTEC Next Generation Mark Scheme Template

IT Unit 1 June 2023

Question Number	Answer	Mark
1ai	Any two from: Web authoring software (1) Desktop publishing software (1) Presentation software (1) Graphics software (1) Multimedia (1) Accept brand names Accept any other relevant software/phrasing/wording.	2

Question Number	Answer	Mark
1aii	Award up to two marks for each of two linked explanations, from: • Enhanced appearance using page layouts (1) because you can arrange/rearrange and manipulate text and graphics with precision to fit into the available space (1) • Document templates available online (1) which will improve the appearance and create a professional document customised to suit the target audience (1) • Can import data from other packages (1) that can be embedded or linked and updated automatically (1) • Available tools such as spell checker, grammar checker, thesaurus (1) to ensure the document is error free (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
1b	Award up to two marks for each of two linked explanations, from: Trojan Horse (1) contains malicious code that carries out actions when executed (1) Worm (1) replicates itself and uses a computer network to spread to other computers (1) Spyware (1) is used to gather information about organisations/people by detecting key presses (1) Adware (1) is a form of spyware that collects information and uses it to display advertisements in a web browser (1) Ransomware (1) restricts access to computer systems by encrypting files on the hard drive (1) Malicious bots (1) can act as spyware and are designed to infect central servers/servers (1) Rootkit (1) enables unauthorised users gaining control of a computer system without being noticed / remain hidden on your computer (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
1c	Award one mark for identification and one additional mark for appropriate expansion. • Ease of use/Intuitiveness (1) ○ Obvious places for users to enter data (1) ○ Have logical flow through the questions to help users navigate (1) ○ Appropriate controls, e.g. buttons/radio buttons/tick boxes/drop down menus (1) ○ Sensible and clear field labels (1) ○ Pop-up instructions to help users (1) ○ Consistent layout/navigation/fonts/colours (1) • Accessibility (1) ○ Provide text alternatives for non-text content (1) ○ Make it available to assistive technologies (1) ○ Allow screen readers (1) ○ Use sufficient contrast to make things easy to see (1) ○ Make text readable/understandable/concise/simple (1)	6

	• Error reduction / accuracy of data (1) ○ Provide ways of reviewing responses (1) ○ Provide ways of undoing/editing (incorrect) responses (1) ○ Validation rules / Error reduction methods e.g. input masks, presence checks, etc. (1) ○ Error messages should give information on how to correct error/proceed (1) • Functionality/ Performance (1) ○ Consider alternative ways of accessing/entering data e.g. onscreen keyboard/attached conventional keyboard (1) ○ Allow for saving on an on-going basis (1) • Compatibility ○ Compatibility with a range of platforms e.g. tablet/smart phone/laptop/PC (1) ○ Compatibility with operating systems/browser e.g. Windows/iOS (1) ○ Consideration of file types/sizes for download (1) Accept any other relevant phrasing/wording	
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Question Number	Indicative content
1d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the ways the teachers can use IT to enhance teaching and learning. <u>Online / eLearning</u> • Materials can be produced in different formats to suit the target audience and learning styles, such as text, videos, animations, etc. • Pupils can access materials where and whenever they choose, which allows a flexible learning approach • Variety of activities can be included to stimulate and motivate learning, such as quizzes, games, tests etc. • Can target specific area of learning to suit the audience, and pupils can revisit materials when required <u>Videoconferencing</u> • Pupils do not have to physically attend lessons if they are off school, the teachers can keep them updated, e.g., due to illness • Cost implications of purchasing specialist computer hardware and software <u>Virtual Learning Environment (VLE)</u> • Can create chat rooms or forums to aid learning, and sharing of expertise • Can operate with small number of pupils and/or more than one class at the same time • Saves travelling to school to access learning, because pupils can work from their own homes via distance learning • Pupils can work at their own pace and access more course materials <u>Interactive Whiteboards (IWB)</u> • Multifunctional device used to teach/demonstrate in the classroom environment • Can engage the pupils using a variety of methods, e.g. sound, video, animation, etc. • Encourages pupil interaction (quizzes) <u>Servers</u> • Used to host e-learning websites and virtual learning environments

	- Learners can access learning resources remotely <u>Personal computers</u> Allows access to e-learning and virtual learning environments for study purposes - Can prepare learning resources
--	---

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-2	- Demonstrates isolated elements of knowledge and understanding - There will be major gaps or omissions - Few of the points made will be relevant to the context in the question - Limited discussion which contains generic points - Little or no consideration of different aspects
Level 2	3-4	- Demonstrates some accurate knowledge and understanding - There will be some gaps or omissions - Some of the points made will be relevant to the context in the question, but the link will not always be clear - Displays a partially developed discussion which considers some different aspects - There will be some consideration of how they interrelate
Level 3	5-6	- Demonstrates mostly accurate and detailed knowledge and understanding - There will be minor gaps or omissions - Most of the points made will be relevant to the context in the question, and there will be clear links - Displays a well-developed and logical discussion which clearly considers a range of different aspects - Clearly shows how they interrelate

Question Number	Answer	Mark
2a	Award one mark for identification and one additional mark for appropriate expansion. Application software A program / set of programs designed for a specific user purpose (1) such as word processing, databases, spreadsheets, etc. (1) Used to carry out end-user tasks (1) without needing to know how the operating system will carry them out (1) System software The interface between the computer and user (1) responsible for interacting with hardware / supporting applications / to hide the complexity (1) Contains the operating system and all utility programs (1) that manage computer resources (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
2bi	Award one mark for identification and one additional mark for appropriate expansion. Access to source code (1) allows customisation to meet Luca's specific needs (1) Luca will be able to work with other like-minded users to 'pinpoint' customisation (1) which is not likely with a large manufacturer (1) Bugs in the code are often spotted and corrected more quickly than with proprietary software (1) as the users are not restricted by large organisation's procedures (1) Can be more secure (1) as the people who are using the software have an inherent need to spot and correct errors quickly / hackers tend to target proprietary software (1) Ongoing support will be free of charge (1) as it is provided by users/community, not the manufacturer (1) More choice of operating system distributions (1) which focus on different user requirements (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
2bii	Award one mark for identification and one additional mark for appropriate expansion. Must ensure OS is compatible with existing application software (1) or he will incur additional costs of replacing it / may have to spend time making the software work with the OS (1) Some (specialised) application software may not run under open-source OS (1) which may make it difficult for Luca and his family to work from home / complete college assessments (1) There may not be as many features included (1) which may make it difficult to carry out specific tasks / costs of additional features (1) Support for the OS relies on the ongoing attention of the users/community (1) which may mean that the support may be sporadic / stop without warning (1) Users may find open-source software harder (1) because they may be less familiar with it / requires greater technical competence (1) Accept any other relevant phrasing/wording	2

Question Number	Answer	Mark
2c(i)	Award one mark for each point up to a maximum of four marks. • (Anti-virus software) monitors activity/scans computer/files (1) • Compares (activity) against a database of definitions (1) • Analyses the code/behaviour of the suspicious program (1) • Determines the nature of the threat (1) • Request actions from the user (1) • Give option to remove the threat (1) • Give option to quarantine the threat (1) • Log user actions to deal with similar future threats (1) Additional Guidance Do not accept 'removes the threat/virus' on its own	4

Question Number	Indicative content
2c(ii)	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers.

Learners to discuss threats to data other than phishing.

Viruses and other malware

Malware:

- Most common types of malware include viruses, trojans, ransomware, bots or botnets, adware, spyware, rootkits, etc.
- Malware software is intentionally designed to cause harm to your computer.
- Malware software is programmed to reproduce, attack, and infect existing files on the system.
- Can target any type of file that the system can execute.

Viruses:

- Executable programs inserted into other programs.
- Remain dormant/hidden and will activate depending on the instructions in the code.
- Replicates itself to gain access to different parts of the computer code
- Transmitted from one computer to another through various means (installation of pirated programs / download of files from unsecure web pages / contaminated removable drives).
- Programmed to mutate and update in order to stay hidden from an antivirus.

Hackers

- Hackers use activities to compromise digital devices for
 - financial gain of stealing credit/debit card numbers or defrauding banking systems, stealing money, request new pins, make purchases, ruin credit, etc.
 - a challenge
 - corporate espionage by stealing product and services information from a competitor to gain advantage
 - a personal attack as revenge on a person/company
 - to cause disruption to someone's computer/network systems.
- Hackers can:
 - Hijack usernames and passwords
 - Sell information to other parties illegally
- Hackers are curious and enjoy trying out new techniques and ways to invade a computer system.
- They use scientific methods to be accurate when performing hacking jobs.

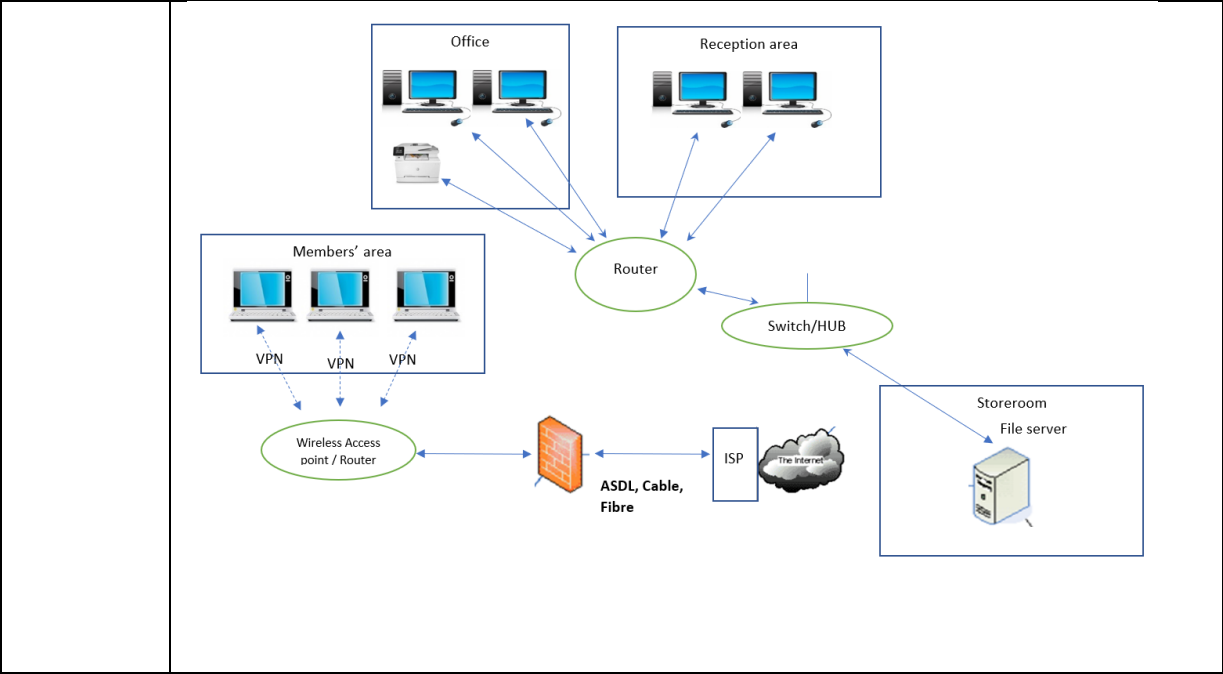
Accidental damage

- This is a situation where data is accidentally lost or damage
- An employee can forget to save a file / delete a file by accident
- Errors when inputting data can cause processing mistakes
- Errors in software can cause the programs to crash and data can be lost
- Power surges/failures can cause data loss.

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4–6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7–8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
3a	A diagram of a potential system / integration of IT systems and connections used to meet the requirements of the scenario i.e. the home working and entertainment system. Example response – note this is indicative only. Credit should be given for a diagram that meets the requirements of the given problem. Diagram may include: Devices: Store room: • Server Reception: • 2 desktop PCs Office: • 2 desktop PCs • 1 printer Members' area: • 3 laptops Staff • 3 tablets Connection types to be used Server - wired Reception desktop PCs – wired Office desktop PCs – wired Office printer – wired Members' area laptops - wireless Staff tablets - wireless Annotations:



Level	Descriptor
0 marks	No rewardable material
1-2 marks	Diagram provides partial coverage of devices. Limited connections are shown Diagram includes some annotations of the connection types used.
3-4 marks	Diagram provides coverage of mostly appropriate devices. Connections are shown which meet most requirements of the scenario. Diagram includes some annotations of connection types used to meet the majority of the requirements of the scenario, but these may not always be the most efficient.
5-6 marks	Diagram provides thorough coverage of appropriate device which fully meet the requirements of the scenario. Connection are shown which fully meet the requirements of the scenario. Diagram includes detailed and accurate annotations of devices and connection types used to meet the requirements of the scenario.

Question Number	Answer	Mark
3b	Award two marks for each point up to a maximum of six marks Real-time (1) - Monitors external events and processes data as soon as it enters the computer system (1) - Allows the system to respond immediately (1) - Used mostly in embedded devices and automated systems (1) Single-user, single task (1) - Only one user would be able to access the PC at any one time (1) - Can only run one task at a time (1) - Used mostly in older mobile phones, PDAs (1) Multi-user (1) - Allows multiple users to use a computer system and its resources at the same time (1) - Used by servers, supercomputers and mainframes (1)	6

	Accept any other relevant phrasing/wording	
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Question Number	Answer	Mark
3c	Award one mark for each point up to a maximum of four marks • Bowser requests a secure page (https://). • Web server sends a public key with its certificate (once secure transaction is initiated) • Web browser checks the certificate is from a trusted party (and is still valid) • Web browser encrypts a random symmetric encryption key using the public key • Sends it to the server with the encrypted URL and other encrypted http data. • Using the private key, the web server decrypts the symmetric encryption key (decrypting the URL and http data). • Web server sends back the requested html document, http data encrypted with the symmetric key. • Web browser decrypts the http data and html document using the symmetric key • Displays the information. Accept any other relevant phrasing/wording	4

Question Number	Indicative content
3d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners will discuss factors that should be taken into consideration when choosing a network (other than purchase cost and security). User experience (ease of use, performance, availability, accessibility) • Ease of Use how easy is it for each user to connect to the network? will users need individual login details? • Performance speed the network will transfer data will the network be able to upload / download media files quickly (for use in training sessions)? limited bandwidth in data transmission method would mean slow data transfer speeds, which can result in logging in, saving and loading files taking a long time firewalls can slow down data transfer (due to checking for threats) • Accessibility will the network be suitable for someone with a disability? will the network be suitable for someone with no knowledge of IT? User needs • Staff training needs (initial & ongoing) how can the leisure centre minimise the effect on the staff? cost of training (in house or outsourced) timing (when will the training be scheduled) • What will the network be used for? will the staff need to access and share media files (for the training sessions)? Specifications • Is the network data transfer speed suitable? • Does the network have maximum bandwidth (for data transfer requirements)? • Is the power consumption suitable? • Is functionality suitable for the leisure centre's purpose? Connectivity • How will the network be connected together, e.g., wired or wireless? • Dependent on user needs for speed, capacity and reliability Efficiency

	• How effective is the network at allowing the staff at the leisure centre to undertake their work? • Wireless connectivity will allow the staff more portability due to less cabling (for the tablets/laptops) slower data transfer speeds • Wired connectivity will increase security, stability and faster data transfer speeds staff will be restricted to their workstations Compatibility • Will the network be compatible with all devices that are to be connected? • Will the tablets (or any smartphones) be able to connect using the wireless connection? • Other systems in use by staff, e.g., portable devices need to be checked to ensure compatibility with the new system • Will desktop PCs need wireless networking cards (in case cabling fails) Implementation (timescales, testing, downtime) • Timescales how long will the installation take? Can it be scheduled for a time when the leisure centre is less busy? is there a target completion date? if an external company is installing the system, is there a penalty if they do not complete on time? if Craig's IT staff are installing the system might other tasks get in the way of the work and increase the time it takes to install? will there be a delay between selecting and implementing wireless connection (ISP/broadband provider)? • Testing has testing been planned thoroughly? they will need to ensure the server/operating system will work with/be suitable for current IT needs must ensure new system will work on all platforms stability software/hardware compatibility if an external company is installing the system, is there a penalty if they do not complete on time? • Downtime how long will the staff be unable to access the network services? is there a contingency plan in place for any potential or unplanned downtime? Productivity • Improve productivity / efficiency with new system
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	• Will staff be able to complete their work quickly and efficiently • Network downtime might have a big impact on productivity, staff would not be able to access resources while system is down
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Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the advantages and disadvantages of different processes and implications for protecting data and systems. File Permissions • Can set file access permissions to suit the user and what they can do with them, this will prevent users from causing damage/deletion of patient data • Protects IT systems from threats, hacking and accidental damage, which can harm and might cause loss of patient data • Read Only, files can be opened, viewed, but not edited, which will prevent users from harming files • Read/Write, files can be opened, viewed and edited, which will limit what users can do with files • Full Control, files can be opened, viewed, edited, modified and deleted, which will allow authorised users full access to patient data Access Levels • Can assign access levels depending on the user profile login, which will prevent users from causing damage to systems • Access levels define the software and data users can access, which will prevent users from causing harm to system functionality • Administrator access, users can access all files with full control, which will allow the administrator to monitor network usage Backup and Recovery Procedures • Regular backups should be taken to reduce the impact of loss of data • Full (backup) will undertake a complete backup of all data, and will provide full protection, but takes time to backup and requires large storage media to store the backups • Incremental (backup) will only backup any data that has been changed since the previous backup. This type of backup is quicker and storage media requires less storage space (than full backup) • Differential (backup) is a mixture of full and incremental. The health centre would take a full backup once a week, and incremental backups every day after. • The health centre needs to put a plan in place to restore data as quickly as possible, because dealing with private sensitive data would cause issues for them Physical Access Controls • Access cards can be used by users so they can be scanned to unlock rooms. Each card is encoded so they work on certain doors which gives a level of security for each user. Could cause a problem if a card was lost or stolen.

	• Keypad access control requires the user to press a passcode to gain entry to a room. Could cause issues if the passcode is not kept secret (passed on to another user, written down, viewed over shoulder). • Biometric access control scans biometric data to gain access to a room. This will be unique to each individual, e.g., fingerprint, iris, facial recognition. Can be costly to the health centre to implement.
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Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5-7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8-10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners evaluate the advantages and disadvantages to the health centre of the use of social media in the health care sector Advantages - Effective form of communication that gives medical practitioners tools to share / discuss / promote health awareness with patients more quickly - Medical alerts can be updated and shared quickly to raise awareness about health problems, e.g., diabetes, allergies, etc. - Medical information/alerts can be shared globally/ to other health care workers - Easier to collaborate with other health care workers to share research/experiences - Easier / quicker to communicate with patients, due to being housebound / transport issues, etc.) - Cost effective marketing for health care workers / patients - Platform to showcase activities and accomplishments, using testimonials to show success - Attract healthcare professionals - Potential patients can search information about Seren from past reviews to choose the right health centre to suit their needs - Ease of use, because most patients are already familiar with social media, and have social media access Disadvantages - Security risks / privacy could be a problem if private/confidential data was leaked/shared - Lack of control of what is being posted, once posted on the Internet it stays there forever - Time consuming to keep social media updated - Reluctance to use a public platform for discussing symptoms/problems - Some patients might prefer face to face contact / human interaction - Some patients might not have access to social media / technology - Some patients might not know how to use technology - Cost implications: - if training is required to enable staff to make use of full functionality, e.g., security, collaboration tools - Patients might find alternative health centres which would cause loss of income for Seren.

Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited evaluation which contains generic or descriptive points • Little or no conclusion
Level 2	5-8	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed evaluation which considers some different competing points • There may be a conclusion which is partially supported
Level 3	9-12	• Demonstrates mostly accurate and detailed knowledge and understanding • There may be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical evaluation which clearly considers different aspects and competing points in detail • There is a conclusion that is supported

**Pearson BTEC Level 3 Nationals Extended Certificate,
Foundation Diploma, Diploma, Extended Diploma**

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Learner Registration Number

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Time 2 hours

Paper
reference

31760H

Information Technology

UNIT 1: Information Technology Systems

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

Answer ALL questions. Write your answers in the spaces provided.

- 1** David is a wedding photographer who takes and edits photographs.

He uses several portable devices including a digital camera, tablet, mobile phone and laptop.

David uses cloud storage for his photographs so that he can access his photographs anywhere with an internet connection.

- (a) Identify **two other** reasons why David uses cloud storage for his photographs.

(2)

1

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David is concerned that he will not be able to access his photographs if there is no internet connection.

- (b) Identify **two other** concerns David may have about using cloud storage.

(2)

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- (c) David has received the following text message that he believes is an attempt at SMS phishing.

OE2T: We were unable to process your latest bil. In order to avoid fees, update your biling information immediately via:

<HTTPS://OE2T-update.ref0170.com/?OE2=2>

- (i) Identify **two** reasons that may have led David to believe that this is a phishing text.

(2)

1

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2

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- (ii) Identify **two** ways that David can protect himself against phishing texts.

(2)

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- (4)

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(ii) Explain **two** benefits to David of using a blog to show the wedding photographs. (4)

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Using a blog is one way David can communicate and interact with online communities.

- (e) Discuss the features of other relevant platforms he could use to upload and show his wedding photographs.

(8)

(Total for Question 1 = 24 marks)



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QUESTION 2 BEGINS ON THE NEXT PAGE.



P 6 7 9 6 6 A 0 7 2 4

- 2 Jamal is the manager of a sports centre, which currently offers training facilities for a variety of sports.

Jamal wants to ensure that the centre is meeting the needs of the local community.

He has decided to gather the opinions of the local community by collecting data from individuals.

- (a) Explain **two** benefits of using primary data to gather the opinions of the local community.

(4)

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- (b) Jamal has decided to collect some of the data using a questionnaire.
Figure 1 shows part of the questionnaire.

We would like to know a little bit about you.

1. Do you currently use the sports centre?

- ☐ Yes
- ☐ No

2. If the answer to question 1 was **Yes**, please tell us the sports you take part in.

- ☐ Football
- ☐ Tennis
- ☐ Squash
- ☐ Badminton
- ☐ Netball

3. If the answer to question 1 was **No**, please tell us what would encourage you to use the sport centre.

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Figure 1

The questionnaire includes multiple-choice and open response questions.

- (i) Give **two** reasons for using multiple-choice questions in a questionnaire.

(2)

1

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(ii) Describe how open question responses are extracted to present findings.

(2)

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(c) Jamal has decided to use online questionnaires rather than hard copies.

Explain **two** benefits to Jamal of gathering data using online questionnaires rather than hard copy questionnaires.

(4)

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(d) Jamal uses questionnaires to collect data from the community.

Explain **one other** method that Jamal could use to collect data from the community.

(2)

(e) Jamal needs to post his questionnaire online.

Discuss the factors, other than question types, that Jamal should consider when producing the user interface for his questionnaire.

(8)



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(Total for Question 2 = 22 marks)



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QUESTION 3 BEGINS ON THE NEXT PAGE.



3 GFC designs and sells computing hardware and software via high street stores.

The company holds stock in each store with additional stock held in a central warehouse.

In-store stock levels are recorded on a database on the in-store server.

Warehouse stock levels are recorded on a database on the company server.

A sales terminal is connected to the in-store local area network (LAN).

The in-store LAN uses the company virtual private network (VPN) to access the company server.

When an item is requested the in-store stock levels are checked.

If the item is not available in-store, stock is checked on the warehouse database.

The status of the stock is sent back to the sales terminal.

(a) Draw a diagram to show the integration of systems used in this process.

The diagram should include:

- devices, systems and connections that could be used
- the flow of data through the system
- annotations indicating data, devices, systems and connections.



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Use this space to draw your diagram for Q3a.

(6)



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The support staff use the company VPN to access the company server.

(4)

[illegible]

- (c) Staff who work remotely must take steps to ensure the security of company data on their devices.

One precaution is to install a firewall on the device.

Describe how a firewall ensures the security of data on a device.

(4)

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- (d) GFC keeps up to date with emerging technologies and wants to introduce robots in its warehouse.

Discuss the implications for GFC of using robots in the warehouse.

(8)



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(Total for Question 3 = 22 marks)



P 6 7 9 6 6 A 0 1 9 2 4

- 4 LN-Educ is a UK charitable organisation providing teaching resources for schools. It wants to offer new online resources that include courses and teaching materials.

LN-Educ's website will provide access for students to use and interact with the new online resources.

The organisation is aware that security of data is important, but there are other implications to consider.

- (a) Discuss the impact and implications, other than security, that the organisation should consider so students can access and use the new online resources.

(10)



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- (b) LN-Educ does not charge for the courses and is keen to attract students from overseas.

However, the organisation is aware that not all overseas students will be able to take advantage of the courses.

Analyse the moral and ethical factors that may restrict access to the courses, and the implications of this for overseas students.

You should consider:

- globalisation
- freedom of speech and censorship
- unequal access to information technology.

(12)

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(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 90 MARKS



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Mark Scheme (Results)

January 2023

Pearson BTEC Nationals
In Information Technology (31760H)
Unit 1: Information Technology Systems

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January 2023

Publications Code 31760H_2301_MS

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Unit 1: Information Technology Systems

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if a candidate's response is not worthy of credit according to the mark scheme.
- Where some judgment is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt about applying the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Phonetic spelling should be accepted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

BTEC Next Generation Mark Scheme Template

ICT Unit 1 Jan 2023

Question Number	Answer	Mark
1a	Any two from: • Can download onto any device / synchronise devices (1) • Easier / quicker to share files (1) • (Automatic) backup (is done by cloud storage provider) (1) • Saves space on devices (1) • Cheap(er) to use (David only pays for the storage he needs) (1) • Convenient (do not need additional hardware) (1) DO NOT ACCEPT Easier, quicker on its own Security / safer Accept any other valid response	2

Question Number	Answer	Mark
1b	Any two from: May worry about: • security / virus (protection) / unauthorised access (1) • losing his files (1) • overload his devices (automatic syncing) (1) • cost (expense once free storage is used up) (1) • online servers could go down (1) • data breach / leak (if photographs get leaked online) (1) Accept 'Hackers' as unauthorised access Accept any other valid response	2

Question Number	Answer	Mark
1c(i)	Any two from: • Incorrect spelling (1) • (Legitimate companies) would not ask for personal / billing details (1) • Urgency ('immediately' in the text is suspicious) (1) • Message makes you fearful of consequences of not responding (1) • There is no 'personalisation' in the text (1) • URL reference / hyperlink (1) Accept any other valid response	2

Question Number	Answer	Mark
1c(ii)	Any two from: • do not click any links / phone numbers (in the message) (1) • do not respond (to the text message) (1) • delete texts (1) • check authenticity of the message (1) • report all phishing attacks (to the appropriate organisation) (1) • message filtering / junk (1) • block (number) (1) • update software (automatically) (1) Accept any other valid response DO NOT ACCEPT: Install anti-virus	2

Question Number	Answer	Mark
1d(i)	Award one mark for each point up to a maximum of four marks. • choose blogging platform / host (1) • install blogging software (1) • create an account / log into his account (1) • choose a blog name (1) • choose a domain extension/name (1) • design blog layout / choose a template (theme) (1) • create / write blog (1) • upload photos (1)	4

	publish / post blog (1) Accept any other valid response	
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Question Number	Answer	Mark
1d(ii)	Award up to two marks for each of two linked explanations, such as: Engagement with readers (1) who can provide online feedback (1) Increase visibility of the posts / new leads / more customers (1) by adding (wedding related) hashtags (1) Tell a story/process about his services (1) so that he can share/demonstrate techniques (1) Cheaper way (to show wedding photographs) (1) rather than the cost of printing to take to wedding conventions (1) Accept any other valid response	4

Question Number	Answer
1e	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the features of other relevant platforms he could use upload and show his wedding photographs. Social media (accept brand names, e.g. Facebook, Instagram, etc.) • location tagging • private/group messaging • create groups • share/like photographs/videos of the weddings • use hashtags to target specific demographics • provides statistics on how many times a photograph has been viewed, shared or commented on • 'Go Live' feature • 'Reporting' feature • accessibility features (e.g. zoom feature) Microblog • short messages that are shared online (text, images, videos, hyperlinks, etc.) • can post (microblogs) on social media sites Vlog (accept brand names, e.g. YouTube) • live streaming • video sharing • links • comments sections • subscribe facilities • 3D videos • 360-degree feature • annotations can be added to videos Wiki • can limit access to a group • simultaneous sharing of documents • can include section tabs/links between pages Instant messaging (IM) / Messaging apps (WhatsApp, etc.) • send (wedding) images to clients (individuals / groups) • send (wedding) videos to clients (individuals / groups) • communicate in real/near real time • notifications/alerts when a message has been received

	• send content using a mobile device • can access chat history • create a chat room with clients Forums • can personalise to ensure content is relevant to audience • share / like (wedding) images and videos • create user groups • email notifications • private / group messaging • group / single sign on facility • can set up security / privacy options
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Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be expanded • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer	Mark
2a	Award one mark for identification and one additional mark for appropriate expansion. (Data collected) is more accurate / valid / reliable / relevant (1) because it is gathered from a specific audience (1) (Data collected) is up to date (1) because it is collected at the time of the research (1) High level of control (1) because Jamal has selected the demographic/questions (1) Accept any other valid response DO NOT ACCEPT: Gathering data from the local community – references the question.	4

Question Number	Answer	Mark
2b(i)	Award one mark for each point up to a maximum of 2 marks. • limits choice (1) • makes analysis easier/quicker (1) • precise/expected responses (1) • easy to input into analytical software (1) • more likely to respond (1) • easily customised / replicated / modified (1) Accept any other valid response Do not award easier/quicker on its own.	2

Question Number	Answer	Mark
2b(ii)	Award one mark for each point up to a maximum of 2 marks. • Look through the data/responses (1) • Identify key / repeating / common themes (1) • Code/tag (to make searchable/countable) (1) • Collate frequency of response (1) Accept any other valid response	2

Question Number	Answer	Mark
2c	Award one mark for each identification and one additional mark for each appropriate expansion. Up to a maximum of four marks. - Improved productivity / easier/quicker to access the results (1) because they are automatically collated / Jamal does not have to physically collate the data (1) - Easier/quicker to collect the data (1) because it is automatically stored electronically (1) - Greater flexibility (1) because he can update the questionnaire to suit his needs (1) - Lower costs (1) because no need for postage / printing hard copies (1) - More accurate (1) because it eliminates Jamal's data entry errors (1) - Higher completion rate (1) because most people could prefer to complete online questionnaires / could be more accessible online (rather than hard copies) (1) - Quicker distribution (1) because it can all be done electronically (1) Accept any other valid response DO NOT ACCEPT Responses relating to the environment, the question is about benefits to Jamal.	4

Question Number	Answer	Mark
2d	Award one mark for identification and one additional mark for appropriate expansion. - survey (1) because they are anonymous (1) - focus groups (1) which allows Jamal to have collective/shared responses (1) - interview (1) which enables Jamal to ask follow on questions (1) Accept any other valid response DO NOT ACCEPT: Forum / poll / observations	2

Question Number	Answer
2e	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss other factors that should be considered when posting a questionnaire online. Ease of Use • obvious places for users to enter data • logical flow through the questions to help users navigate • pop-up instructions to help users • consistency layout • appropriate/consistent fonts/colours • consistent navigation techniques Accessibility • provide text alternatives for non-text content • allow 'zoom' feature • available to assistive technologies • provide contrast to make things easy to see • make text readable/understandable/concise/simple Error reduction • minimise data entry requirements • appropriate controls, e.g. buttons/radio buttons/tick boxes/ drop down menus • include data validation techniques Intuitiveness • use standard icons • use standard navigation methods Functionality • is all the functionality keyboard accessible? • consider other ways of accessing, e.g. touch screen • provide ways of 'undoing' incorrect responses Performance • ensure minimal lag between submission and next question Compatibility • consider different platforms, e.g. tablets/PC • consider different operating systems, e.g. windows/IOS

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
3a	A diagram of a potential system / integration of IT systems and connections used to meet the requirements of the scenario i.e. the GFC systems Example response – note this is indicative only. <pre> graph TD ST[Sales terminal] -- Stock request --> ISS[In-store server] ISS -- Stock availability --> ST ISS -- Stock check --> ISD[(In-store database)] ISD -- Stock availability --> ISS ISS -- VPN --> R1([Router]) R1 -- Stock request --> I[The Internet] I -- Stock availability --> R1 R1 -- VPN --> CS[Company server] CS -- Stock check --> WD[(Warehouse database)] WD -- Stock availability --> CS </pre> Credit should be given for a diagram that meets the requirements of the given problem.

	Diagram may include: Devices: Sales terminal In-store server In-store file server/ database In-store router Company router Company server Company/warehouse file server / database Connections: Sales terminal to in-store server In-store server to router In-store router to company router – VPN Company router to company server Company server to database server/database Data flow: All connections will have two-way data flow Essentially requests for stock going in-store to warehouse Information/results coming back Annotations: Annotations indicating data, devices, and systems.
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Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.	
Level	Descriptor
0 marks	No rewardable material
1-2 marks	Diagram provides partial coverage of appropriate devices. Storage/Data sharing and connection types are used to meet some of the requirements of the scenario. Diagram includes annotations of the data that will be shared among the components of the system.
3-4 marks	Diagram provides coverage of mostly appropriate devices. Storage/data sharing and connection types are used to meet the majority of the requirements of the scenario, but these may not always be the most efficient. Diagram includes mostly accurate annotations of the data that will be shared among the components of the system

5-6 marks	Diagram provides thorough coverage of appropriate devices. Storage/data sharing and connection types are used to fully meet the requirements of the scenario. Diagram includes detailed and accurate annotations of the data that will be shared among the components of the system.
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Question Number	Answer	Mark
3b	Award one mark for each point up to a maximum of four marks. • VPN software used (by the sender) (1) • (the software) encrypts data (1) • sends (encrypted) data to the VPN server (1) • the (encrypted) data is sent over the internet (1) • through a secure connection (1) • (encrypted) tunnel (1) • the (encrypted) data is decrypted (by the VPN server) (1) Accept any other valid response. DO NOT ACCEPT: 'encrypts the connection'	4

Question Number	Answer	Mark
3c	Award one mark for each point up to a maximum of four marks. • the firewall forms a barrier between the device and a network (1) • using either hardware and/or software (1) • which are connected using designated ports (1) • the firewall filters/checks incoming/outgoing traffic (1) • against a set of rules / filter the data / compare to blacklist/whitelist (1) • blocks connections / stops (unauthorised) access to the system/data (1) • It alerts the user to anything it finds suspicious/harmful (1) Accept any other valid response. DO NOT ACCEPT: 'Blocks' on its own	4

Question Number	Answer
3d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Discuss the implications to GFC of using robots in the warehouse. Benefits of using robots in the warehouse Improved efficiency/productivity • can improve performance/productivity because human error can be quite costly for GFC and will lead to customer dissatisfaction • the warehouse will be able to handle larger quantities of goods each day, which will speed up order completion • repetitive tasks can be completed by the robots, which will allow GFC's employees to spend time on other tasks which require critical thinking • production will be more consistent, because factors such as distractions, tiredness, experience, staff turnover could affect employees • robots offer flexibility because they can run continuously (24/7), only stopping for maintenance needs. Reduction in errors • accuracy will increase because robots are programmed to perform same/repeated tasks more accurately, resulting in higher quality products • using robots for routine operations in the warehouse reduces the risk of human mistakes, allowing employees to focus on quality control measures Reduced overheads / running costs • lower operating costs, because robots do not need to work in heated or lit environments • reduced employment costs to GFC, because fewer employees may be needed and robots do not require salaries Warehouse safety / improved working conditions • safety for workers is improved, because robots can undertake dangerous jobs that could put GFC's employee's safety at risk (e.g. working at heights, carrying heavy loads, etc.) • robots can be used to transfer large items safely and without damage • reduces the chances of accidents (falling items, trips, falls, etc.) Customer satisfaction • increased delivery speeds / and decreased human error will give the GFC a competitive edge and increase customer satisfaction

	• using robots in the warehouse GFC can allow workers more time to concentrate on customer relations Drawbacks of using robots in the warehouse Cost • the initial purchasing of the robots can be quite costly in the short-term rather than hiring employees for the warehouse or using other equipment such as forklifts • additional electricity costs will be incurred • specialist technicians are required to undertake ongoing maintenance of the robots which would be expensive for the GFC, because their salaries would be higher • robots require skilled robotic programmers, which would cost GFC substantial money Adaptability • robots don't adapt to changing situations in the workplace, and are difficult to re-program to accommodate the changes • errors will be shown if identifying labels are unreadable or misplaced on the products which will require assistance from human employees GFC employees • employees might be concerned about losing their jobs, because robots are introduced to replace some employees Different types of warehouse robot Picking • can be used to move around the aisles collecting/depositing items on shelves • picking robots reduce order processing times Sorting • can be used to identify items and deposit in the correct storage area Packing • can be used to speed up packing operations • can calculate the package size using the weight and dimensions of the item Transportation • can be used to decrease the physical demand of moving items around the warehouse Replenishment • robots can be programmed to automatically replenish stock/send alerts when levels drop below the minimum threshold Do Not Accept 'work breaks' for costs
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Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learner to discuss the impact and implications, other than security, that the organisation should consider so students can access and use the online resources. Ease of use/intuitiveness • must be easy to use/intuitive – users' familiarity/knowledge of using websites for education purposes will range widely and they may not keep using the site if they encounter problems/difficulties • obvious places for students to enter data when required • update the page regularly with current content/resources • consistency layout/navigation/fonts/colours • have logical flow through the pages/subjects/resources to help users navigate • appropriate controls, e.g. buttons/radio buttons/tick boxes/drop down menus • pop up instructions to help users Accessibility • LN-Educ should incorporate accessibility features, to ensure inclusive access to content/resources for all students • provide text alternatives for non-text content • make it available to assistive technologies • incorporate features that will allow ease of use for visually impaired, e.g. use sufficient contrast to make things easy to see, make content adaptable, suitable fonts/colours to assist readability • incorporate features for hearing impaired, e.g. captions and alternatives for audio and video content • features for specific disabilities, e.g. do not use content that causes seizures • language options Performance • no lag when interacting with the website • compatibility with different web browsers Availability • consider different platforms, e.g. tablets/PCs/laptops • 24/7 access to students Cost • updating and maintaining the website, courses and materials

	• May need to employ extra IT staff initially and/or long term Testing • testing of the online resources need to be planned thoroughly • ensure the website is tested for stability and software/hardware compatibility User support • onsite help/call desk facilities
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Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5–7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8–10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners analyse the factors which may hinder access to the courses and the implications for overseas students. Learners should consider: • Globalisation • Freedom of speech and censorship • Unequal access to information technology <u>Globalisation</u> Globalisation implies that all nations/ countries/states can use the advances in telecommunication systems to interact, regardless of geographical location. However, some nations/ countries/states within countries do not benefit from globalisation for a variety of reasons: • the electricity infrastructure may be inadequate • broadband connections may be non-existent or intermittent • may only have access to potentially slower wireless infrastructures • students may need to go to a 'central' location, such as a school, to access the internet, there may only be one connection point for all students to share • the hardware/software available may be outdated – often second hand • may not have skilled IT staff to help them develop skills <u>Freedom of speech and censorship</u> Not all governments allow individuals to have complete freedom to use the internet as they wish. They may block: • the internet • specific search engines • searches/queries on specific topics • social media sites <u>Unequal access to information technology</u> Due to the factors given above access to IT is unequal. The implications of this for overseas students are: Education: • Limited access to course materials may mean that students are disadvantaged in terms of the quality and accuracy of their knowledge. • Lack of IT infrastructure may mean that they cannot follow these courses

	Employment: • They may not be able to develop desirable skills such as technical or programming skills which could lead to better paid jobs • Lack of infrastructure may mean they cannot access specific technologies that allow them to apply for employment or to be able to subscribe to working styles (remote working etc.) • Lack of access to information may restrict them from the application process or mean they are not aware of potential jobs. Financial: • A lack of 'public' infrastructure may mean that access to internet and other technologies is restricted to those that can afford expensive mobile technology • An inability for an area to support high speed online services, may mean that the economy of that area suffers as they cannot compete with other countries in a global market.
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Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-4	• Demonstrates isolated knowledge and understanding, there be major gaps or omissions • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question • Limited analysis which contains generic assertions rather than interrelationships or linkages
Level 2	5-8	• Demonstrates some accurate knowledge and understanding, with few minor omissions/any gaps or omissions are minor • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained
Level 3	9-12	• Demonstrates mostly accurate and thorough/detailed knowledge and understanding • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question • Displays a well-developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner



Please check the examination details below before entering your candidate information

Candidate surname					Other names					
Pearson BTEC Level 3 Nationals Extended Certificate, Foundation Diploma, Diploma, Extended Diploma	Centre Number					Learner Registration Number				
Monday 20 January 2020										
Morning (Time: 2 hours)					Paper Reference 31760H					
Information Technology										
Unit 1: Information Technology Systems										
You do not need any other materials.								Total Marks		

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

- 1** Fulmill is an inner-city suburb. Fulmill Community Group (FCG) has been set up to encourage local people to become involved in the community.

FCG produces a newsletter providing local news and information. The newsletter is currently printed to be collected from local shops. FCG is considering replacing the printed copy with an online version.

- (a) (i) Give **two** drawbacks for members of the community of replacing the printed copy with an online version.

(2)

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- (ii) Give **two** benefits for members of the community of replacing the printed copy with an online version.

(2)

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- (b) Kerry is a member of FCG. She has a BITMAP image that she wants to include in the FCG newsletter.

She is going to send the image as an email attachment. FCG has asked Kerry to convert the file into JPEG format.

- (i) Explain **one** benefit to FCG of converting the file from BITMAP (.bmp) format to JPEG (.jpg) format.

(2)

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- (ii) Explain **one** drawback to FCG of converting the file from BITMAP (.bmp) format to JPEG (.jpg) format.

(2)

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P 5 9 7 9 8 A 0 3 2 4

- (c) Two protocols used to control the data transmission of emails are Simple Mail Transfer Protocol (SMTP) and Internet Message Access Protocol (IMAP).

Explain the roles of SMTP and IMAP in the transmission of emails.

(4)

SMTP

IMAP

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(d) Kerry has antivirus software installed on her computer.

Kerry keeps the antivirus software up to date.

Describe how keeping the antivirus software up to date reduces the risks of viruses on the computer.

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(e) FCG is considering setting up an online community on a social media site.

Discuss how FCG can ensure that the social media site provides a positive user experience for the online community.

(8)

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QUESTION 2 BEGINS ON THE NEXT PAGE.



2 Bassett's college has a local area network (LAN) that can be accessed by students on-site and remotely.

- (a) Access to the server room is restricted by a keypad entry system. A four-digit code is issued to authorised users on a weekly basis.

The college is considering changing to an ID card entry system.

Explain **two** ways in which an ID card entry system makes access to the server room more secure than a keypad entry system.

(4)

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(b) Students gain access to the college network with usernames and passwords.

Describe the process used by the system to authenticate passwords and allow access to the system.

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- (d) Many students use personal laptops to access the college network. This can have implications for the security of data stored on the network.

Discuss systems and procedures that the college and students could use to keep data on the network secure, when students access the network using their personal laptops.

(8)

(Total for Question 2 = 20 marks)



P 5 9 7 9 8 A 0 1 1 2 4

3 Joe runs a small company. The company is moving to a purpose-built office.

Joe has asked an IT expert for advice about suitable IT systems. The IT expert has proposed several alternative solutions.

One solution is to set up a local area network (LAN) that will provide:

- wired connections for five PCs
- a file server
- a backup server
- two networked printers
- secure internet access
- wireless access for laptops and other mobile devices.

(a) Draw a diagram to show the setup of the LAN.

The diagram should include:

- the devices used by the system
- the connections between the devices
- annotations indicating the devices and the types of connections.

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Use this space to draw your diagram for Q3a.

(6)



(b) The proposed LAN has five PCs with wired connections.

Explain **two** implications of using a wired rather than a wireless connection.

(4)

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- (c) A network operating system will be installed on the file server. One role of a network operating system is to manage security.

Describe **two other** roles of a network operating system.

(6)

1

2



P 5 9 7 9 8 A 0 1 5 2 4

- (d) One other proposal is to use cloud storage for backup rather than using an on-site server.

Discuss the implications for the company of using cloud storage for backup rather than an on-site server.

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(Total for Question 3 = 24 marks)



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QUESTION 4 BEGINS ON THE NEXT PAGE.



4 GBA provides financial services and advice to clients.

The company is relocating to a smaller office.

Following the move, GBA will offer some employees the option to work from home.

(a) Discuss how technology enables employees to work effectively from home.

You should consider factors affecting:

- access to systems and data
- communication
- data security.

(10)

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- (b) As part of the move, GBA is considering updating all of its IT systems, some of which are over five years old.

Evaluate the advantages and disadvantages of updating all of its IT systems. You should consider the impact on:

- GBA
- the employees.

Your answer should include a recommendation for GBA on whether to proceed with the update of its IT systems.

(12)

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(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 90 MARKS





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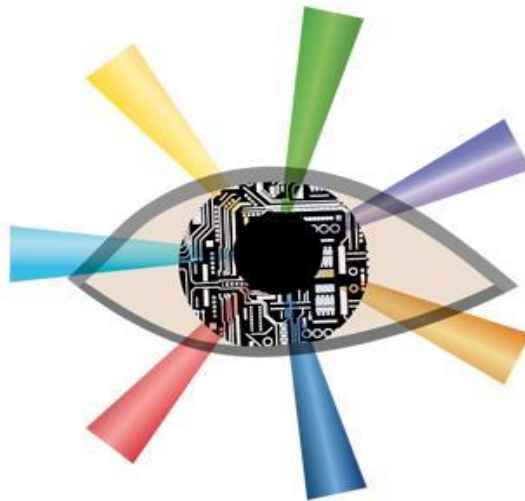
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Mark Scheme (Results) January 2020

BTEC Level 3 Nationals



Information Technology

Unit 1: Information Technology Systems 31760H

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January 2020

Publications Code 31760H_2001_MS

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Unit 1: Information Technology Systems

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if a candidate's response is not worthy of credit according to the mark scheme.
- Where some judgment is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt about applying the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Phonetic spelling should be accepted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

BTEC Next Generation Mark Scheme

IT Unit 1 January 2020

Question Number	Answer	Mark
1ai	Award one mark for each drawback up to a maximum of two marks: Not everyone will: • have the necessary IT literacy skills (1) • have access to IT facilities / internet (1) • be able to afford IT equipment (1) • if site went down no access to online version (1) • might be more difficult to read if accessing on a mobile device (1) Accept any other valid response.	2

Question Number	Answer	Mark
1aai	Award one mark for each benefit up to a maximum of two marks: • improves availability for those who are housebound / opens up availability to a wider/younger audience / more inclusive (1) • e-newsletter can be viewed wherever/whenever to suit members of the community (1) • enables readers to share stories/items (with friends/followers) (1) • digital features can be incorporated (e.g. videos, links, social media sharing buttons) (1) • quicker updates on events • accessibility features for visually impaired (zoom feature) • benefits to the environment (1) - if the response clearly refers to 'environmental issues for the community, e.g. because people don't cause litter in the community by throwing away the newsletter / printed copy'. Note Do not award for general environmental issues, e.g. less printing and less paper required, because this refers to the company producing the printed copy. Accept any other valid response.	2

Question Number	Answer	Mark
1bi	Award up to two marks a linked explanation, such as: - the file size will be reduced (1) - the image will be compressed (1) - the e-newsletter will render more quickly (1) - it will be faster to download/ upload the image/email (1) - it is less likely that it will be rejected by the FCG's' email provider (1) Accept any other valid response.	2

Question Number	Answer	Mark
1bii	Award up to two marks a linked explanation, such as: the quality of the JPEG images will not be as good (as bitmap) (1) and the JPEG may appear pixelated/ distorted (1) Accept any other valid response.	2

Question Number	Answer	Mark
1c	Award up to two marks for each of two linked explanations, such as: SMTP – manages outgoing emails / used for sending emails (1) from server to server (1) until it reaches its recipients (mail) server (1) IMAP – used for receiving emails/ incoming emails (1) stores emails on server (1) allowing access from multiple devices/synchronises email across devices (1) Accept any other valid response.	4

Question Number	Answer	Mark
1d	Award one mark for each point up to a maximum of four marks: - to recognise and handle new virus strains (1) - New viruses are constantly being created / virus threat is constantly evolving (1) - Content is compared against (an existing) database of virus definitions (1) - If new virus is not in existing list, it will not be detected (1) - antivirus software developers constantly monitor the threat landscape (1) - antivirus software developers will update the definitions database when any new virus is found (1) Accept any other valid response.	4

Question Number	Indicative content
1e	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss how FCG can ensure that the social media site provides a positive user experience for the online community. The page / group /website must provide a 'good' user experience: • must be easy to use / intuitive – users' familiarity / knowledge of using social networking will range widely. Users will not be active/keep using if they encounter problems/ difficulties • must be accessible on different platforms • meet the needs of all users. Accessibility – may be users with special accessibility requirements, must ensure these are built in • update the page regularly with current content • availability of the site (no downtime) The online community must be set up so that: • a moderating team oversees that 'rules of behaviour' are set and adhered to • users are aware of what is/isn't acceptable to post to avoid any offence / inappropriate material being posted • the privacy of users is given high priority. Need to consider who is going to be given access. Can be 'closed group' or extended so that anyone can access • users are aware of potential dangers of using social media • take measures to ensure the security of users' data. Users must be confident that the security has been given high priority when setting up Users should: • communicate with each other in real time • raise issues about issues/events in local community • add in links to websites /blogs etc which may be helpful

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4–6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7–8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer	Mark
2a	Award up to two marks for each of two linked explanations, such as: • Unauthorised users can see / guess the four-digit number as it is being entered into the key pad (1) an ID card system does not provide this visual detail, reducing the risk of unauthorised access (1) • Authorised entry can only be gained by physically having the card (1) therefore unauthorised users would need to steal/copy/borrow the card (1) • ID cards have in built 'technology' (1) making them difficult to copy/requiring specialist equipment to enable copying (1) • Authorised users should/will be aware of the risks/outcomes of lending cards to others (1) reducing the likelihood of them passing on their cards (1) • If ID card is lost / stolen the college can deactivate the card (1) preventing anyone finding the card gaining access (1) • An ID card is unique to each individual (1) can monitor who accesses the server room / which provides evidence of whose ID entered the server room (1) • An ID card often includes a photo of the person to whom it has been issued (1), providing a means of checking that the user of the card is the person authorised to do so (1) Accept any other valid response.	4

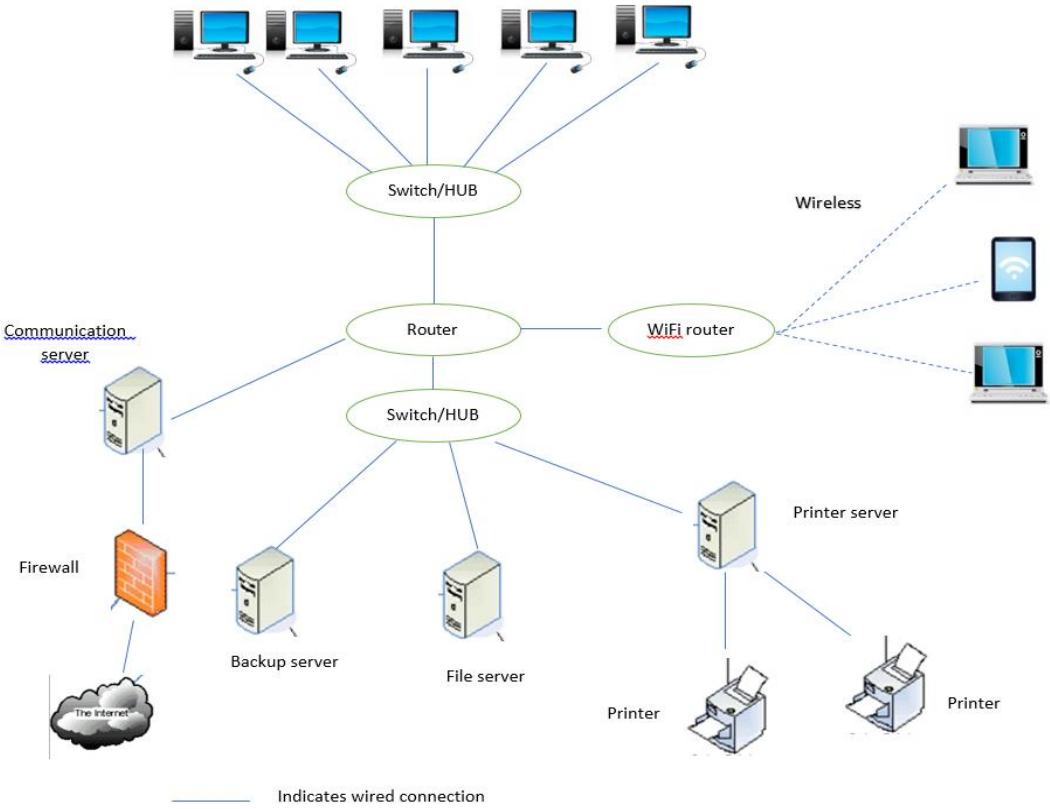
Question Number	Answer	Mark
2b	Award one mark for each point up to a maximum of four marks: • Database of usernames and passwords created (1) • on entry password matched the username (1) • access allowed if they are the same/match (1) • re-enters password if no match / access denied (1) • system allows limited number of attempts (1) • account locked when unsuccessful attempts exceeds attempts allowed (1) Accept any other valid response.	4

Question Number	Answer	Mark
2c	Award up to two marks for each of two linked explanations, such as: • A shared area can be set up in which documents can be saved / stored centrally with updated filenames (1) allowing others to access/edit /comment on the documents (1) • Version control can be maintained (1) to ensure that all users are working on the same version of the document (1) • Manage groups (1) so only invited members of the group will have access to the group resources (1) • (As they are working on the same network), users will have access to the same software (version) (1) no compatibility issues (1) • Will be able to use real-time communication software (virtual meetings /chat facilities) (1) without having to be in the same physical location (1) • co-authoring features provided by applications running on the network (1) which will enable students to work together in real time (1) • students can set up a group calendar/project plan for the project (1) which will help them monitor progress and meet deadlines (1) Accept any other valid response.	4

Question Number	Indicative content
2d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Students discuss systems and procedures that the college and students could use to keep data on the network secure, when accessing it using their personal laptops. Procedures to prevent threats to storage data • Acceptable use of IT policies eg network wifi login • Up to date antivirus/ antimalware on college network and laptops • Ban/ encrypt USBs/ portable media • Firewall on the network • Health check for laptops • Regular backups Procedures for preventing unauthorised access to the personal laptops • Firewall installed on the laptops in addition to the network to prevent unauthorised access • Password protection / multi-factor authentication on the laptop, prevents unauthorised access Procedures for preventing unauthorised access to the network via personal laptop • Set up appropriate access levels on network for staff/students • File permissions on network for staff/students • Techniques for preventing other threats to data stored on the laptops being transferred to college network e.g. Anti-virus / malware software installed on laptop • VPN Physical methods to protect data stored on personal laptops • Physical access to laptops. Students must take responsibility for 'looking after' the laptops e.g. locked away overnight, never left anywhere others can access them • Students should use external storage devices with caution, applying the same sort of physical access controls to storage devices e.g. USB drives / external hard drives

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
3a	Requirements: Devices – 5 PCs, file server, backup server, two networked printers Wired connections – PCs/server/backup server/printers Wireless access – laptops / mobile devices Secure internet access Annotations – types of connections and devices Note: Do not penalise for not having a separate printer server or a communication server.  <pre> graph TD Router((Router)) --- S1((Switch/HUB)) Router --- S2((Switch/HUB)) Router --- WiFiRouter((WiFi router)) S1 --- PC1[PC] S1 --- PC2[PC] S1 --- PC3[PC] S1 --- PC4[PC] S1 --- PC5[PC] S2 --- CommServer[Communication server] S2 --- Firewall[Firewall] S2 --- BackupServer[Backup server] S2 --- FileServer[File server] S2 --- PrinterServer[Printer server] PrinterServer --- P1[Printer] PrinterServer --- P2[Printer] WiFiRouter -. Wireless .-> Laptop[Laptop] WiFiRouter -. Wireless .-> Smartphone[Smartphone] WiFiRouter -. Wireless .-> Tablet[Tablet] Firewall --- Internet((The Internet)) </pre> — Indicates wired connection

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	Diagram provides partial coverage of appropriate devices. Storage/Data sharing and connection types are used to meet some of the requirements of the scenario. Diagram includes annotations of the data that will be shared among the components of the system.
Level 2	3–4	Diagram provides coverage of mostly appropriate devices. Storage/data sharing and connection types are used to meet the majority of the requirements of the scenario, but these may not always be the most efficient. Diagram includes mostly accurate annotations of the data that will be shared among the components of the system
Level 3	5–6	Diagram provides thorough coverage of appropriate device. Storage/data sharing and connection types are used to fully meet the requirements of the scenario. Diagram includes detailed and accurate annotations of the data that will be shared among the components of the system.

Question Number	Answer	Mark
3b	Award up to two marks for each of two linked explanations, such as: • Signal is constant/reliable/stable (1) not affected by interference/walls/other devices (1) • Signal does not deteriorate over distance (as much) (1) reduce the need for boosters (1) • More secure (1) because it is not as easily intercepted/ no need to set up encryption/WPA/WEK etc (1) • Restrict movement / no mobility / cables have to be laid (1) PC must stay where the connection point is (1) • Wired connections are faster (transmission speeds) than wireless connections (1) allowing large files to download faster (1) • Health & Safety risks (1) trailing cables / tripping hazards etc (1) Do not award for cost. Accept any other valid response.	4

Question Number	Answer	Mark
3c	Award up to three marks for each of two descriptions: Networking (1) Manages communication (protocols) within the network and externally (1) to govern and control data transmission (1) as well as managing and maintaining the network using utility programmes (1) Memory management (1) Allocates space/memory to files/programs/clipboard items that are running/open (1) decides how much each process will get process will get memory and at what time (1) monitors each memory locations status i.e. free/allocated (1) Multi-tasking (1) Allocates each task a slice of processor time (1) keeps a track of where the user is in each task (1) and goes from one to the other without losing information (1) Device drivers (1) Driver receives signals from/ communicates with hardware (1) translates signals so software (OS & applications) can understand (1) when a signal is received from hardware OS it is held by a buffer / instruction put into a queue until resources available (1)	6

Question Number	Indicative content
3d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the implications for the company of using cloud storage for backup rather than a local server. Cloud based backup • Stored / managed by third party • Stored offsite • cost must be weighed up • pay only for what is used – if need increased storage, can pay for extra. • environmental issues • must ensure have security protocols in place which comply with the GDPR • giving a third party access to secure data introduces extra risk • there may be security concerns with using online services, as these are often targeted by hackers because of their 'high profile' nature • backup performance is dependent on the quality and availability of an internet connection • can be set up to automatically back up / sync with mobile devices, ensuring that latest information is available to those that need it. This also improves security – if any device lost/stolen data will be available on server • likely to have more robust disaster recovery in place • access remotely • Large storage capacity Onsite server • company has complete control, in particular control over security of data • cost to be weighed up – cost of server / may need to employ IT staff to run the system • if extra capacity is needed will need to purchase additional server • no reliance on internet connection • cannot be accessed remotely / cannot auto sync with mobile devices unless additional systems put in place e.g. VPN • offsite storage of backups should be considered • natural disasters, e.g. fire / flood

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects

Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss how technology has impacted on the ability of employees to work from home. Access to systems and data: • Networking – users can access company networks using VPNs and remote desktop applications. Gives Financial Advisors access to all of the files they will need/would have if working in the office. Also ensures data security • If the company use cloud storage will also give access to files and ensure synchronisation of files so that everyone in the organisation is working of the same files. Users will not need to worry about backing up files. • Cloud computing can be used so that everyone is working on the same software versions, reducing compatibility issues. Home workers don't have to worry about buying / installing up-to-date software on PCs/ laptops • May be able to take advantage of remote desktop technologies which will provide a similar experience to working in the office Communication: • A range of online communities /VoIP / video conferencing can be used / set up to ensure that the Financial Advisors don't feel isolated / maintain communication • Data transmission broadband / cellular widespread with increasingly good/reliable/stable connections. • Financial advisers will be able to communicate with clients over secure cellular networks Data security: • Security is a major issue as Financial Advisors will be dealing with highly sensitive client data. Security is enhanced by using secure internet connections. • Encryption can be used to secure data during transmission. • Threats to data stored on systems can be reduced by using sophisticated anti-virus, firewalls, encryption • Access levels to protect sensitive data

Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5–7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8–10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners evaluate the advantages and disadvantages of updating all its IT systems. Cost / benefit • GBA will incur large initial costs for IT systems • May have to employ extra IT staff, initially and/or long term • Cost of staff training • May be able to re-coup some finance from selling on used equipment Disposal of old equipment • May need to bring in experts (which will be costly) to ensure all data is removed or storage media destroyed • Time and cost to dispose of the hardware in an environmentally friendly way Implementation • Direct change over could impact the operations of GBA if the new systems doesn't work and the current system is obsolete • Phased current staff could get used to the new system in phases • Parallel, downtime would be held to a minimum with the current systems still running while the new ones are being implemented Integration • Other systems in use by staff e.g. portable devices need to be checked to ensure compatibility with the new office system • Files could be lost/ corrupted during transfer Productivity • Loss of productivity during implementation • Improve productivity / efficiency with updated systems Working practices • Changes to IT systems might have an impact on employees working from home eg updating mobile technology Staff training needs (initial & ongoing) / User support • How can GBA minimise the effects on staff • Cost • Timing • In house or outsourced? Testing • Must ensure that new system will work on all platforms • Will need to test this prior to move as far as possible • Need to ensure new systems will work with/be suitable for current IT needs

	Security • Data during changeover • Setting up new systems • Up-to-date security software
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Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited evaluation which contains generic or descriptive points • Little or no conclusion
Level 2	5–8	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed evaluation which considers some different competing points • There may be a conclusion which is partially supported
Level 3	9–12	• Demonstrates mostly accurate and detailed knowledge and understanding • There may be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical evaluation which clearly considers different aspects and competing points in detail • There is a conclusion that is supported



Llywodraeth Cymru
Welsh Assembly Government



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Candidate surname					Other names					
Pearson BTEC Level 3 Nationals Extended Certificate, Foundation Diploma, Diploma, Extended Diploma	Centre Number					Learner Registration Number				
Thursday 6 June 2019										
Morning (Time: 2 hours)					Paper Reference 31760H					
Information Technology										
Unit 1: Information Technology Systems										
You do not need any other materials.								Total Marks		

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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- (4)

2

(b) Explain **two** drawbacks of using a wiki to collaborate with other students.

(4)

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P 6 1 5 8 9 A 0 3 2 4

- (c) Wigmore College should ensure that computer workstations are designed to minimise health risks.

Explain **two** factors the college should consider when designing computer workstations in order to minimise health risks.

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(d) Edie has bought a laptop to use when working at home.

She has been advised that she should set up a firewall to protect the data on her laptop.

Describe how a firewall works to protect data on the laptop.

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- (e) Edie is considering purchasing a suite of office productivity applications for the laptop rather than purchasing individual applications.

Discuss the implications of this purchase.

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(Total for Question 1 = 24 marks)



- 2 Richard is a football trainer. He has set up a company offering training and competitions for footballers of all abilities, aged 16 and over.

He would like to create a website that will:

- advertise training facilities
- advertise competitions
- collect personal details of players
- allow players to book training events and enter competitions.

The website must be accessible to all users.

- (a) Describe **two** accessibility features that should be included in the website.

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QUESTION 2(B) BEGINS ON THE NEXT PAGE.



(b) Personal details of players will be entered into an online data collection form.

Figure 1 shows part of a completed data collection form.

First name	Kelvin
Last name	Barbaros
Date of birth	17/07/2002
Telephone number	07700 900729
Address 1	88 Torrent Place
Post code	SR8 7GC
Email	kbarbaros@bgesal.org

Figure 1

It is important that the data collected is error free.

(i) Validation is used to minimise errors in the data collected.

Explain what is meant by the term 'validation'.

(2)



(ii) Describe a validation rule that could be used for each of the following:

(4)

Date of birth

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Email address

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(iii) Describe **one other** technique that could be used to minimise errors in data entry.

(2)

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- (c) The online booking system should allow players to book training events and enter competitions.

The system should be easy to use and accessible for all users.

Discuss the factors, other than ease of use and accessibility, that Richard should consider when creating the online booking system.

(8)

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(Total for Question 2 = 20 marks)



3 ERS Trading sells sports goods.

It sells goods online and in high street stores.

The company offers a 'click-and-collect' service for online customers.

Customers using the 'click-and-collect' service nominate a store for collection.

On receipt of an online order the availability of stock in the warehouse is checked.

Items, that are not available, are ordered from suppliers.

Available items are despatched to the nominated store.

Items not collected by the customer within 14 days are returned to the warehouse.

Emails are sent to the customer when:

- the item is despatched to the nominated store
- the item is ready for collection in-store
- items have not been collected by the customer within 7 days.

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(a) Draw a flow chart to show the process for 'click-and-collect' ordering.

(6)



P 6 1 5 8 9 A 0 1 5 2 4

(b) Philippe is the manager of one of the larger ERS Trading stores and is concerned about the amount of time customers spend queuing to pay for goods.

- (i) Describe **one** way Philippe could use data modelling to help him solve the problem.

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(ii) Explain **two** implications of using a data model to solve this problem.

(4)

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P 6 1 5 8 9 A 0 1 7 2 4

(c) Online customers pay using debit cards. It is essential that their payment details are kept secure.

Describe the protocols used to govern data transmission for secure payment systems.

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4 Frederick's Agency is a chain of accountants. Some accountants are based in a main office, others work from home.

The agency has a range of clients, including small and medium-sized businesses.

The main office has a server-based Local Area Network (LAN), which can be accessed via Ethernet cabled PCs or remotely via a company Virtual Private Network (VPN).

The server provides access to all application software and company data.

Company data is currently backed-up to a cloud-based storage system.

The agency is pleased with the cloud-based back-up and is considering using cloud storage and computing for all its IT systems. The agency believes this will be cost effective.

- (a) Evaluate the cost implications for Frederick's Agency of a move to cloud storage and computing.

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(b) The moral and ethical factors of the use of IT systems has implications for the running of Frederick's Agency.

Discuss the implications for the agency and its staff of the moral and ethical factors of the use of information technology.

(10)

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(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 90 MARKS



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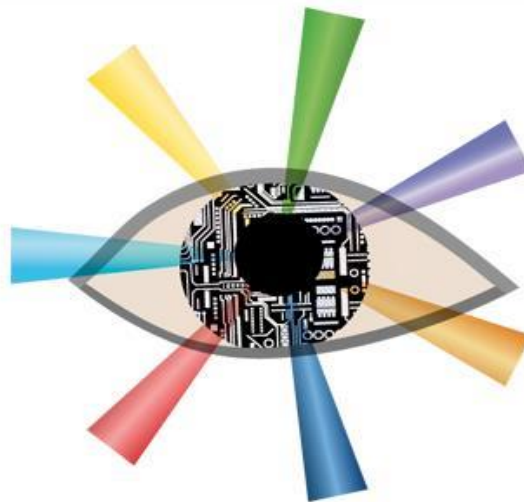
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Mark Scheme (Results) June 2019

BTEC Level 3 Nationals



Information Technology

Unit 1: Information Technology Systems 31760H

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June 2019

Publications Code 31760H_1906_MS

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Unit 1: Information Technology Systems

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if a candidate's response is not worthy of credit according to the mark scheme.
- Where some judgment is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt about applying the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Phonetic spelling should be accepted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Mark
1a	Award one mark for identification of a benefit and one additional mark for appropriate expansion. Can limit access to the group / can set up the system to only accept invited participants (1) ensures that the group have control over contributions (1) Editable/ customizable/ updated/ an embed/ upload documents (e.g. spreadsheets / calendars / images / videos / web links) (1) by only the members of the group (1) Simultaneously sharing of documents (1) reduces time sending document to all of the group / everyone see the same document (1) Ease of use (1) easy to set up / don't require advanced editing skills / knowledge of computer programming (1) Simple to navigate (1) can include section tabs / links between multiple pages (1) Generally unlimited size (1) group can create complex / media-rich wikis (1) WYSIWYG (what you see is what you get) (1) all of the group see the same page (1) Version control (1) users can refer back to previous versions/ can recover previous edited information/ data (1) Additional guidance: identification and expansion marks maybe transposed. Accept any other relevant phrasing/wording.	4

Question Number	Answer	Mark
1b	Award one mark for identification of a drawback and one additional mark for appropriate expansion. Tracking changes can be difficult (1) may be difficult to quickly identify recently edited content / identify content to be updated (1) Not all edits will be agreeable/ acceptable to all of the group (1) need to establish a protocol for accepting changes (1) Open to spam / vandalism (1) if not managed properly (1) Unreliable information (1) if not verified/ validated (for accuracy) (1) Limited formatting options /features eg fonts/colours/styles (1) which may be detrimental to the overall output (1) The software may be difficult for non-technical users (1) which means all members of the groups may not be able to contribute (1) A wiki doesn't provide analytic (1) which may impact on the research project (1) Additional guidance: identification and expansion marks maybe transposed. Accept any other relevant phrasing/wording.	4

Question Number	Answer	Mark
1c	Award one mark for the identification of a factor and one additional mark for the appropriate expansion to a maximum of four marks. (Provision of) ergonomic furniture / adjustable chairs / ergonomic computer equipment / designed to support users (1) to minimise back/posture problems (1) Positioning of furniture/monitors (1) to ensure appropriate height/distance etc between elements e.g. chairs and desks/monitors (1) Flexibility of layout/ furniture (1) to ensure that different users can adjust the workstation to suit individual needs (1) Ensure minimal reflected light on screens (1) position screens appropriately/provide/use window blinds (1) Cables/wires should be stored securely / tied up (behind PCs) (1) to avoid risk of tripping/injury by students (1)	4

	Adjustable height / tilting monitors (1) so that users can minimise neck problems / adjust to individual eye-level (1) Provision of ergonomic mice /ergonomic keyboards (1) to reduce RSI (1) Additional guidance: identification and expansion marks maybe transposed. Accept any other valid response.	
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Question Number	Answer	Mark
1d	Award one mark per each of the appropriate expansion up to a maximum of four marks. • Uses software and/or hardware (1) • Monitors incoming/outgoing files/traffic/packets (1) • Checks the content by enforcing a set of rules/ database known viruses (1) • Identifies suspect traffic (1) • Decides which (files/traffic packets) are allowed to enter or leave (1) Accept any other valid response.	4

Question Number	Indicative content
1e	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the implications, of purchasing a suite of office productivity applications for the laptop rather than purchasing individual applications. Advantages • Generally a suite will be cheaper than buying individual application packages • Commonality between terminology/layout/menu systems, makes it easier to learn/use subsequent programs in the suite • The 'Help' function will be common to all applications • Ability to share data/link objects between programs, which improves efficiency • Customised settings can be applied to all products in the suite/instead of to each individual program, saving time/improving efficiency • Integration of features between programs in the suite which will improve efficiency • Software update only need to be applied once which will save time • Updates are less likely to cause issues with other programs in the suite as the update is tested within the suite and not just on a standalone product • Once the correct version is purchased, there will be no compatibility issues with the laptop and individual apps Disadvantages • Each component is likely to have less features than a stand-alone program which may limit what Edie can do and this may cause problems when collaborating with others • May have components that are not required but are included in the price, may be cheaper to buy only the necessary ones • Storage space may be an issue. If the suite includes unwanted applications does this take up more space than if only required applications are purchased • May not be fully compatible with the college/other students software which may cause problems when collaborating

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer	Mark
2a	Award one mark for the identification of a feature and one additional mark for the appropriate expansion to a maximum of four marks. Incorporate features that will allow ease of use for visually impaired (1) • text alternatives for non-text content (1) • zoom facility (1) • make it available to assistive technologies (1) • use sufficient contrast to make things easy to see (1) • make content adaptable (1) • appropriate fonts/colours to assist readability (1) Incorporate features for hearing impaired (1) • captions and alternatives for audio and video content (1) Features for specific disabilities • e.g. epilepsy (1) do not use content that causes seizures (1) General features to aid usability (1) • make all functionality keyboard accessible (1) • give users enough time to read and use content (1) • provide adequate navigation guidance to help users find content (1) • consistency of layout/navigation (1) • make content appear and operate in predictable ways (1) • help users avoid and correct mistakes (1) Additional guidance: Do not give more than one mark for the same identification point. Accept any other valid response.	4

Question Number	Answer	Mark
2bi	Award one mark for identification and one additional mark for appropriate expansion. Identification (Validation is a technique) to ensure that data is acceptable /sensible /reasonable /allowable (1) Expansion • by checking that it follows a given set of rules /format / is within acceptable boundaries (1) • during data input/entry (1) • carried out by computer software (1) Do not award for checking data is accurate/correct/valid Only one expansion mark may be awarded	2

Question Number	Answer	Mark
2bii	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks. Date of birth Format check (1) e.g. dd/mm/yyyy (1) Length check (1) e.g. 10 characters (1) Range check (1) e.g. DoB >= 16 years (1) Presence check (1) e.g. must be entered / is not null (1) Email address Type check (1) e.g. check for string (1) Length check (1) e.g. 20 characters (1) Format check (1) e.g. must contain '@' and '.' (1) Presence check (1) e.g. must be entered / is not null (1) Identification and expansion marks can be transposed. Accept any other valid response.	4

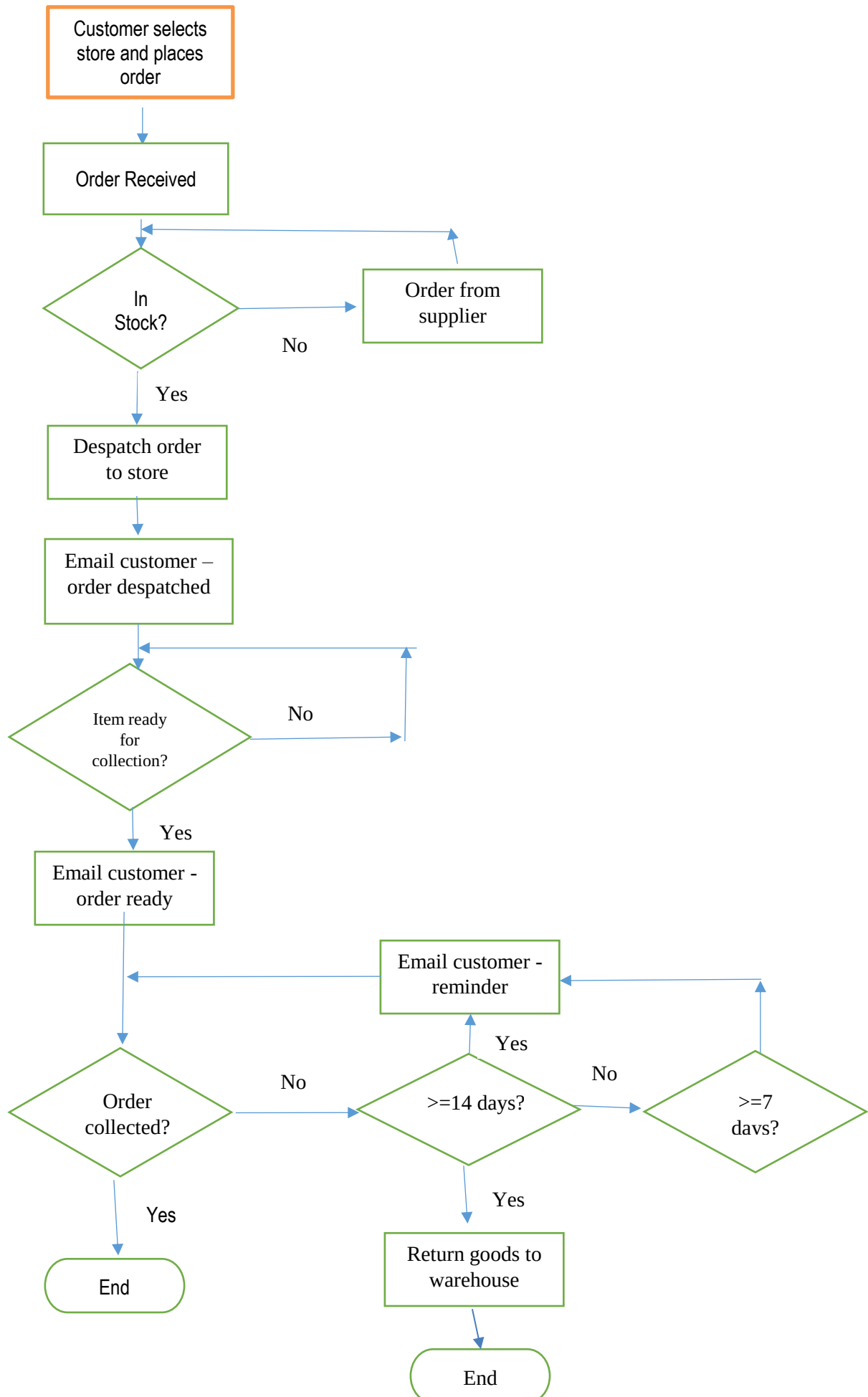
Question Number	Answer	Mark
2biii	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of two marks. Verification checks (1) double entry / checking against original documents/ proof reading/ checks against stored data (1) Accept any other valid response.	2

Question Number	Indicative content
2c	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the factors that Richard should consider when creating the online booking system. Specifications/compatibility • compatibility with database of players (yet to be created) • compatibility with tablet/mobile devices/apps • compatibility with different web browsers Connectivity • to host server • bandwidth available • performance Cost • development cost • hardware costs • ongoing/running costs • training costs Efficiency/productivity • system availability Implementation • timescales • testing Security • data protection • encryption of data when booking via online service Payment systems

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

3a Example of a flowchart



Question Number	Indicative content
3a	A flow chart showing the process for click-and-collect ordering Example response – note this is indicative only. Credit should be given for a flow chart that meets the requirements of the given problem. Flow chart should include logic for: • placing of order • nomination of a store • checking availability of stock in warehouse • ordering from supplier • despatch to store • return of goods to warehouse after 14 days • emails sent when: • item despatched • item ready for collection • items not collected within 7 days

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	Diagram/ Flowchart provides partial coverage of appropriate decisions and shows some logical structure to meet the requirements of the scenario. Diagram/ Flowchart includes annotations to show the results of decision making.
Level 2	3–4	Diagram/ Flowchart provides coverage of mostly appropriate decisions and logical structure to meet the majority of the requirements of the scenario, but the structure may not be the most efficient. Diagram/ Flowchart includes accurate annotations and technical language to show the results of decision making.

Level 3	5-6	Diagram/ Flowchart provides thorough coverage of appropriate decisions and logical structure to fully meet the requirements of the scenario. Diagram/ Flowchart includes accurate annotations and technical language to accurately represent an efficient system.
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Question Number	Answer	Mark
3bi	Award one mark for any of the following: Use spreadsheet software (1) Using inputs – e.g. number of checkouts / number of customers / amount of time a customer spends queueing / time of day / day of the week (1) Data analysis/ trends (1) To produce outputs e.g. busiest times/ number of customers queueing (1) To determine which options minimise/optimize queueing time/outputs (1) Accept any other valid response.	4

Question Number	Answer	Mark
3bii	Award one mark for identification on an implication and one additional mark for appropriate expansion. The variables can be changed (1) to allow comparison of different scenarios (1) Can run in accelerated time (1) don't need a day/week to try out scenarios (1) Cannot (predict) model exactly what will happen/ May be factors which interfere with 'normal' events (1) is only a representation of real situation/ may produce unreliable results (1) Accurate data entry is necessary (1) to produce usable results (1) A model is dependent getting the 'rules' correct (1) to produce usable results (1) Additional guidance: identification and expansion marks maybe transposed. Accept any other valid response.	4

Question Number	Answer	Mark
3c	Award one mark for identification and one additional mark for appropriate description. Hypertext Transfer Protocol Secure (HTTPS) (1) Authenticates the accessed website/ adds a layer of security on the data in transit (1) Secure Socket Layer (SSL)/ Transport Layer Security (TLS) (1) Establishes a secure link between a web server and browser (1)	4

	Encryption (1) Converts data to an encoded form/ scrambled/ cannot be read (if intercepted) (1) Public Key Infrastructure (PKI) (1) Enables an unsecure public network to securely exchange data/ manages digital certificates/ digital signatures/ public keys (1) Allow encryption only once Accept any other valid response.	
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Question Number	Indicative content
3d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners analyse the benefits and drawbacks for customers of the use of transactional data collected through loyalty card schemes. Benefits and Drawbacks depend on the point of view of the candidate. Award marks for justified responses. <u>Benefits</u> • purchase histories can be used for targeted marketing e.g. issuing money-off vouchers for goods frequently purchased, • purchase histories can be used to target advertising, which can benefit customers as 'irrelevant' advertising may be reduced • information may be shared with third parties, which is then used for targeted marketing, which may be beneficial/welcome <u>Drawbacks</u> • Customers may become inundated with special offers/ vouchers • schemes can log where you buy (when buying in store), which can give information on a person's movements/location/routines, leading to privacy concerns • information may be shared with third parties, which are used for targeted marketing, which may be unwanted • there may be concerns over precisely who sees the data / whether data is passed on to third parties • can be used to create a 'profile', including personal details such as income / family status / members of the family • customers may be concerned about the amount of personal information they are asked to provide e.g. 'age' may be collected and customers may not see the relevance • customer may have concerns over how data is protected

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material
Level 1	1-2	• Demonstrates isolated knowledge and understanding, there be major gaps or omissions • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question • Limited analysis which contains generic assertions rather than interrelationships or linkages
Level 2	3-4	• Demonstrates some accurate knowledge and understanding, with few minor omissions/any gaps or omissions are minor • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5-6	• Demonstrates mostly accurate and thorough/detailed knowledge and understanding • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question • Displays a well-developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner

Question Number	Indicative content
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners evaluate the cost implications for Frederick's Agency of a move to cloud storage and computing. Learners provide an evaluation of the decision, stating if they think the decision is appropriate or not. Their evaluation should be supported by relevant points which may include: Using a cloud storage and computing system as an alternative to the current system will bring a range of cost advantages to the agency. In the traditional current system they had to pay for the server and its installation, pay for and install applications and configure the system. There were also a range of ongoing costs relating to maintenance/updating etc. Using a cloud based system the agency don't physically buy a server but access a part of a much larger server which is off-site away from the business. This is likely to be much cheaper in the long run, although there will be costs incurred with running the system. <u>Costs will be saved</u> as there is no need to pay for: • expensive hardware on an ongoing basis – either to update or to add capacity • updating software • ensuring up-to-date security • technical staff to maintain the system / deal with user issues etc • software, including OS and applications. • unused server capacity, which would be fixed under current system, but now can be scaled up and down to fit the needs of the business • access to up-to-date technology, without having to make huge investments • robust disaster recovery, which could be very costly in terms of expertise. • bespoke application packages or other commercially available software, cloud business applications are offered by suppliers and are much cheaper • VPN for home based workers to access server <u>Costs will be incurred for:</u> • access to the new system. May be subscription based or pay as you go model, which will give them authorisation to use the software for a period of time and pay only for the software that they are using • Included in the cost will be a payment for support/staff etc. • server capacity, which will be dependent on what they are using • mobile apps for staff, as required, allowing use on a variety of devices to access software & data

Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material
Level 1	1-4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited evaluation which contains generic or descriptive points • Little or no conclusion
Level 2	5-8	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed evaluation which considers some different competing points • There may be a conclusion which is partially supported
Level 3	9-12	• Demonstrates mostly accurate and detailed knowledge and understanding • There may be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical evaluation which clearly considers different aspects and competing points in detail • There is a conclusion that is supported

Question Number	Indicative content
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the implications for the agency and its staff of the moral and ethical factors of the use of information technology. The company should have policies set in place to guide staff and also to monitor adherence to the policies. <u>Privacy</u> Must ensure the privacy of all details collected/shared online for both staff and clients. For example the workers who mainly work from home may visit clients and complete reports etc after the meeting. This should take place in an environment where the details cannot be 'seen' by other people, e.g. not in public places/train/café etc. Must also ensure the security of the information. The company and staff must be aware of the need to dispose of printed matter securely to avoid details getting into the 'wrong hands'. <u>Environmental</u> The company has responsibility for ensuring the safe disposal of IT equipment. Expensive hardware such as old PCs etc could be reused by other organisations / colleges etc, individual parts can often be recycled. Staff can take an effective part in this by ensuring that smaller items e.g. print cartridges are not thrown into normal waste. Paper – all staff should be aware of the need to reduce paper usage / recycling. <u>Online behaviour and netiquette</u> Each organisation will have its own rules and regulations about acceptable behavior, staff must adhere to these <u>Codes of practice</u> Set up to self-regulate, allowing staff to report any causes for concern to the organisation. Allows the organisation to deal with issues as they see fit in accordance with their own policies. Sets a clear set of common ground rules across the organisation and avoids misunderstandings. It is the responsibility of the staff to familiarise themselves with them. <u>Health and Safety</u> Organisations are morally and legally obliged to look after the Health and Safety of their employees. <u>Changing working practices</u> Availability of technology may result in employees working outside the standard working hours. This may affect work-life balance and health.

Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material
Level 1	1-4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5-7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8-10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate



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